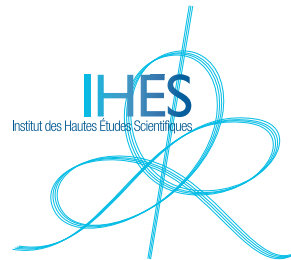


High-Precision Dynamics and Waveform in Two-Body Gravitational Scattering

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Theoretical Developments in Gravitational Wave Physics
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Acronyms

PN Post-Newtonian approximation (**expansion in $1/c$; ie v^2/c^2 and $GM/(c^2 r)$**)

PM Post-Minkowskian approximation (**expansion in G ; ie in $GM/(c^2 b)$**)
and its recent **Worldline EFT avatars**

MPM Multipolar post-Minkowskian approximation

SF Gravitational Self-Force: expansion in m_1/m_2 ,

EOB Effective One-Body Approach

NR Numerical Relativity

EFT various flavours:

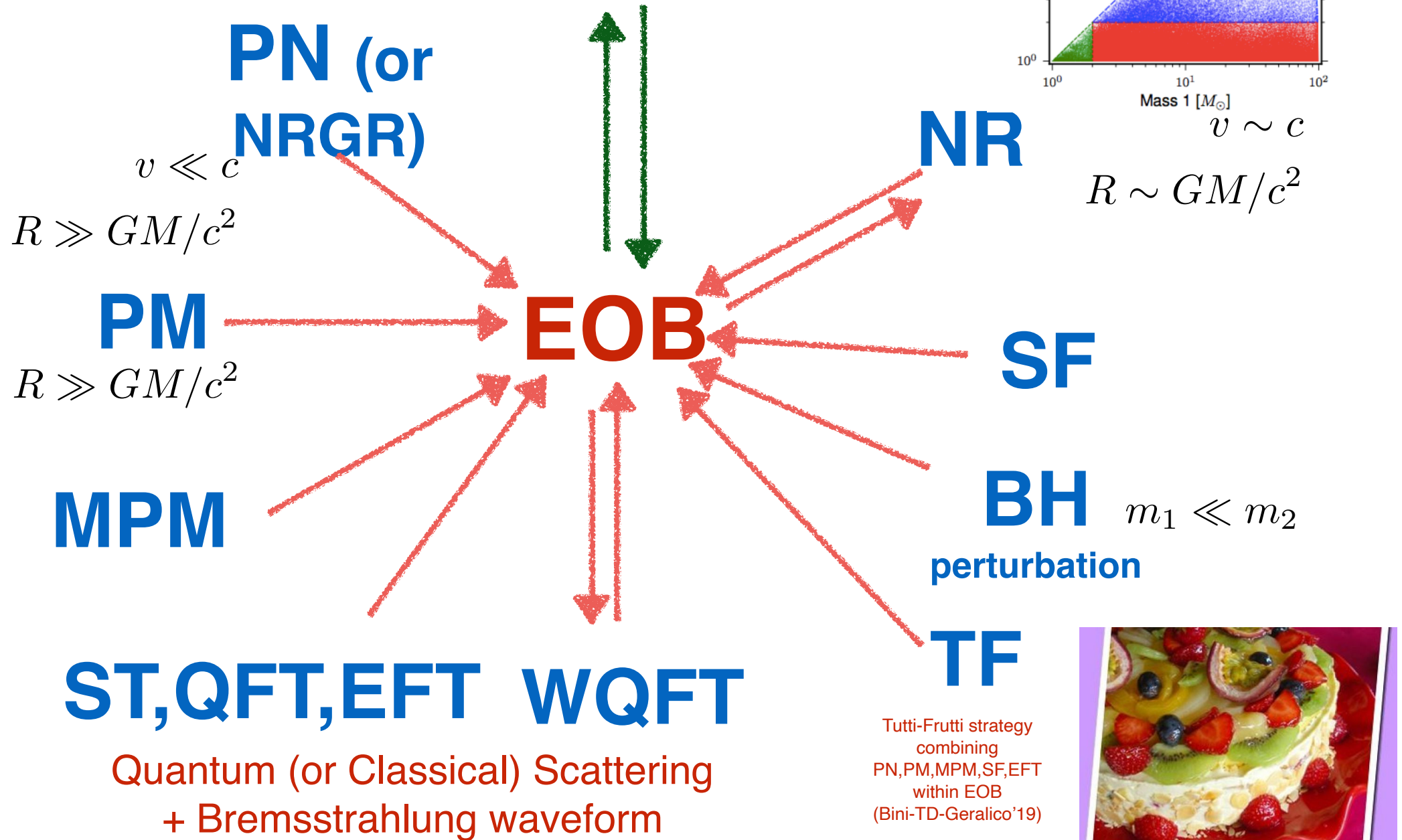
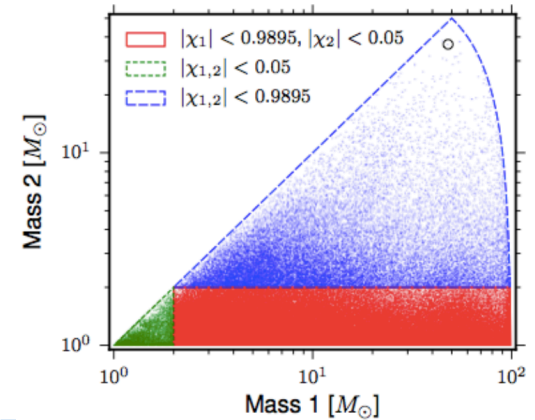
NRGR Effective Field Theory (EFT) à la Goldberger-Rothstein

Classical/Eikonal limit of Quantum scattering amplitude

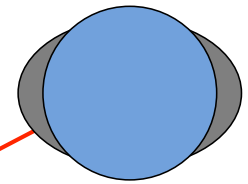
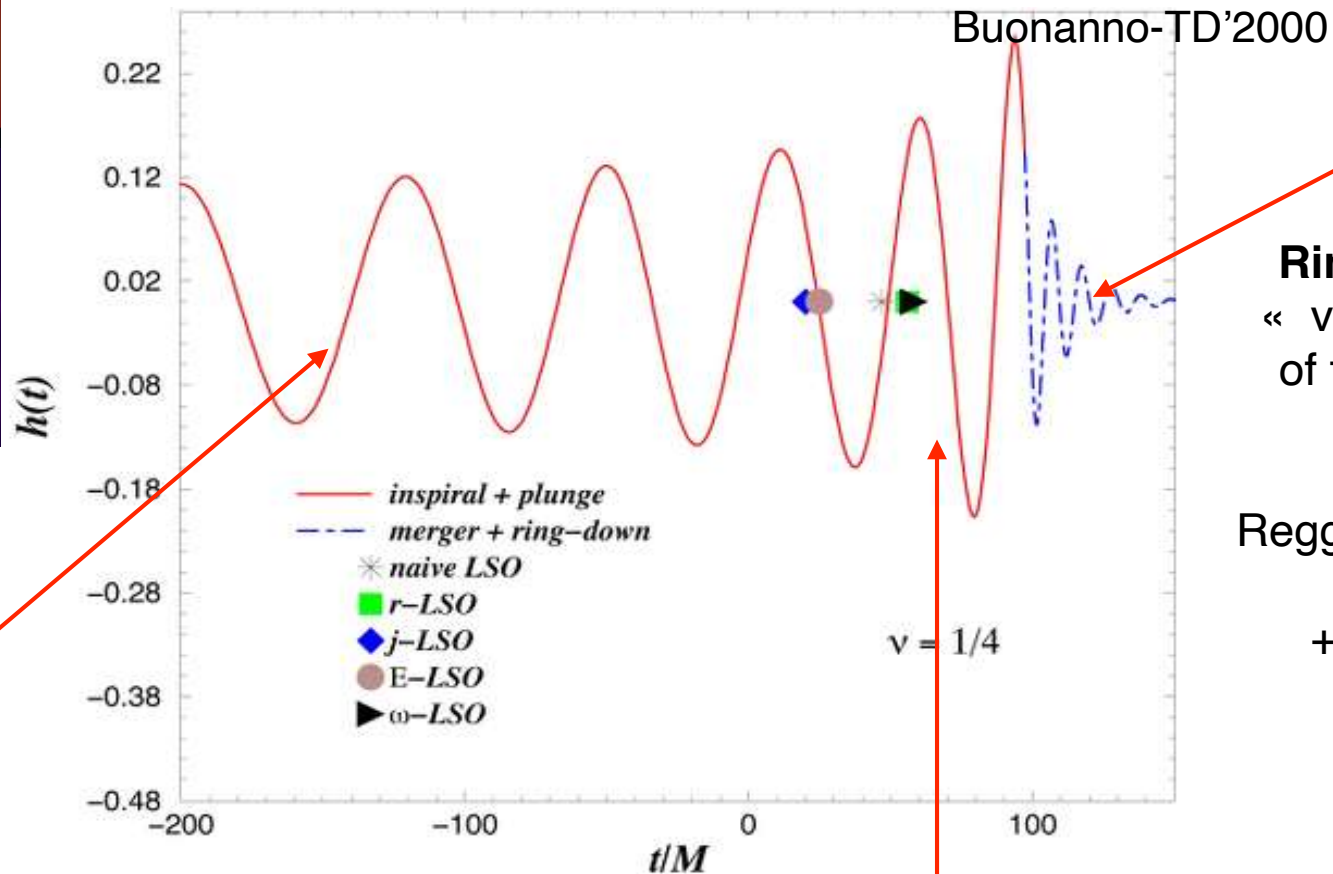
Worldline EFT

TF Tutti Frutti method

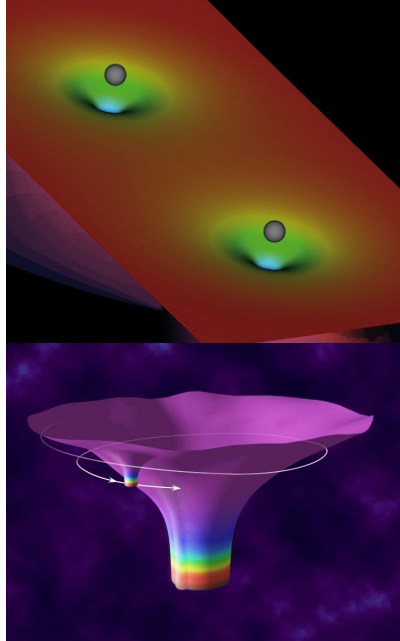
several of LIGO-Virgo-Kagra's
banks of search templates



The Effective One-Body (EOB) approach to the GW signal emitted by the Merger of two Black Holes



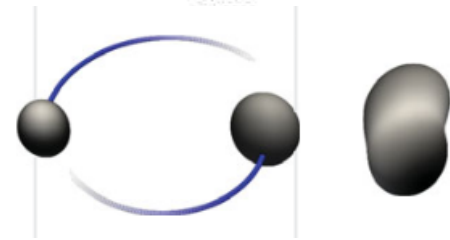
Ringdown (BBH):
 « vibration modes »
 of final BH (QNM);
 perturbation
 of BHs à la
 Regge-Wheeler-Zerilli-
 Teukolsky
 +Vishveshwara



Inspiral:
 perturbative
 computation
 of higher-order
 contributions
 to $E=H$ and F
 (expansion in v^2/c^2
 + tidal polarizability
 + of NS)

Late inspiral, « plunge » and merger:

first estimated by the Effective One-Body method (AB-TD 2000)
 later confirmed and improved by using
 numerical simulations (Pretorius...2005)

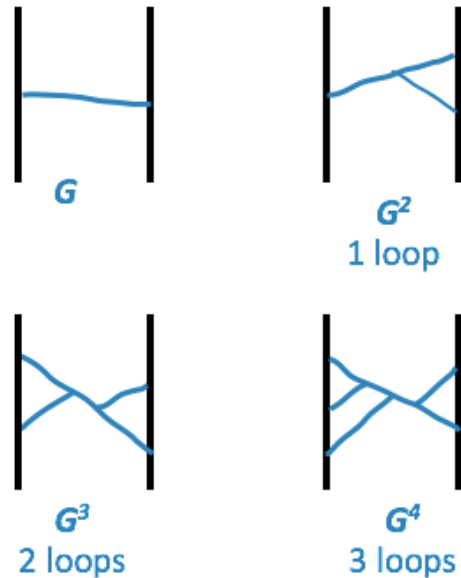


Effective One-Body (EOB) approach: **H + Rad-Reac Force**

Historically rooted in QM: Brezin-Itzykson-ZinnJustin'70
eikonal scattering amplitude+ Wheeler's: 'Think quantum mechanically'

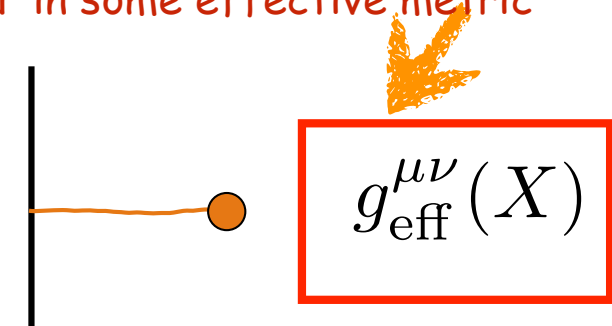


Real 2-body system
(in the c.o.m. frame)



An effective particle of mass μ in some effective metric

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$



mass-shell constraint

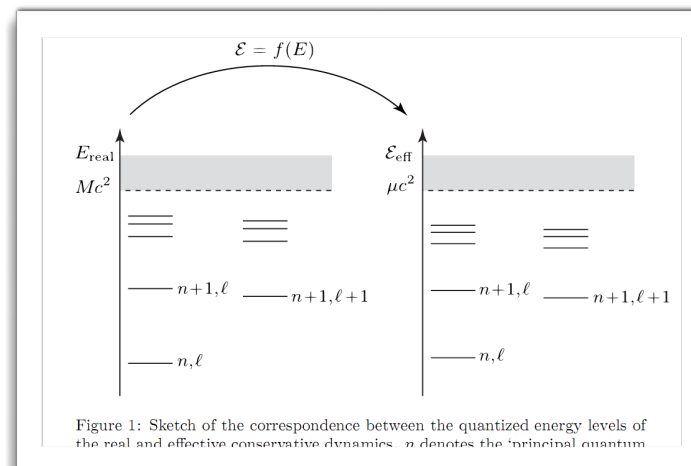
$$0 = g_{\text{eff}}^{\mu\nu}(X) P_\mu P_\nu + \mu^2 + Q(X, P)$$

Level correspondence
in the semi-classical limit:
Bohr-Sommerfeld ->
identification of
quantized action variables

$$J = \ell \hbar = \frac{1}{2\pi} \oint p_\varphi d\varphi$$

$$N = n \hbar = I_r + J$$

$$I_r = \frac{1}{2\pi} \oint p_r dr$$



Crucial energy map

$$\mathcal{E}_{\text{eff}} = \frac{(\mathcal{E}_{\text{real}})^2 - m_1^2 - m_2^2}{2(m_1 + m_2)}$$

as functions of I_r and $I_\varphi = J$

2-body Taylor-expanded 4PN Hamiltonian [DJS, 2014, JS 2015]

$$c^8 H_{4\text{PN}}^{\text{local}}(\mathbf{x}_a, \mathbf{p}_a) = \frac{7(\mathbf{p}_1^2)^5}{256m_1^5} + \frac{Gm_1m_2}{r_{12}} H_{48}(\mathbf{x}_a, \mathbf{p}_a) + \frac{G^2m_1m_2}{r_{12}^2} m_1 H_{46}(\mathbf{x}_a, \mathbf{p}_a) \\ + \frac{G^3m_1m_2}{r_{12}^3} (m_1^2 H_{441}(\mathbf{x}_a, \mathbf{p}_a) + m_1m_2 H_{442}(\mathbf{x}_a, \mathbf{p}_a)) \\ + \frac{G^4m_1m_2}{r_{12}^4} (m_1^3 H_{421}(\mathbf{x}_a, \mathbf{p}_a) + m_1^2m_2 H_{422}(\mathbf{x}_a, \mathbf{p}_a)) \\ + \frac{G^5m_1m_2}{r_{12}^5} H_{40}(\mathbf{x}_a, \mathbf{p}_a) + (1 \leftrightarrow 2), \quad (\text{A3})$$

$$H_{48}(\mathbf{x}_a, \mathbf{p}_a) = \frac{45(\mathbf{p}_1^2)^4}{128m_1^8} - \frac{9(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2(\mathbf{p}_1^2)^2}{64m_1^6m_2^2} + \frac{15(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2(\mathbf{p}_1^2)^3}{64m_1^6m_2^2} - \frac{9(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1^2)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{16m_1^6m_2^2} \\ - \frac{3(\mathbf{p}_1^2)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{32m_1^6m_2^2} + \frac{15(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_1^2)^2\mathbf{p}_2^2}{64m_1^6m_2^2} - \frac{21(\mathbf{p}_1^2)^3\mathbf{p}_2^2}{64m_1^6m_2^2} - \frac{35(\mathbf{n}_{12} \cdot \mathbf{p}_1)^5(\mathbf{n}_{12} \cdot \mathbf{p}_2)^3}{256m_1^5m_2^3} \\ + \frac{25(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)^3\mathbf{p}_1^2}{128m_1^4m_2^3} + \frac{33(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)^3(\mathbf{p}_1^2)^2}{256m_1^4m_2^3} - \frac{85(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{256m_1^4m_2^3} \\ - \frac{45(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{128m_1^3m_2^3} - \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2(\mathbf{p}_1^2)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{256m_1^3m_2^3} + \frac{25(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{64m_1^3m_2^3} \\ + \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{64m_1^3m_2^3} - \frac{3(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^3}{64m_1^3m_2^3} + \frac{3\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^3}{64m_1^3m_2^3} + \frac{55(\mathbf{n}_{12} \cdot \mathbf{p}_1)^5(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_2^2}{256m_1^5m_2^3} \\ - \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_1^2\mathbf{p}_2^2}{128m_1^5m_2^3} - \frac{25(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1^2)^2\mathbf{p}_2^2}{256m_1^5m_2^3} - \frac{23(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4(\mathbf{p}_1 \cdot \mathbf{p}_2)\mathbf{p}_2^2}{256m_1^5m_2^3} \\ + \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{128m_1^3m_2^3} - \frac{7(\mathbf{p}_1^2)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{256m_1^3m_2^3} - \frac{5(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^4\mathbf{p}_1^2}{64m_1^4m_2^4} + \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_2)^4(\mathbf{p}_1^2)^2}{64m_1^4m_2^4} \\ - \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)^3\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{4m_1^4m_2^4} + \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{16m_1^4m_2^4} - \frac{5(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_2^2}{64m_1^4m_2^4} + \frac{21(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_1^2\mathbf{p}_2^2}{64m_1^4m_2^4} \\ - \frac{3(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2(\mathbf{p}_1^2)^2\mathbf{p}_2^2}{32m_1^4m_2^4} - \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1 \cdot \mathbf{p}_2)\mathbf{p}_2^2}{4m_1^4m_2^4} - \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)\mathbf{p}_2^2}{16m_1^4m_2^4} + \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2\mathbf{p}_2^2}{16m_1^4m_2^4} \\ - \frac{\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2\mathbf{p}_2^2}{32m_1^4m_2^4} + \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4(\mathbf{p}_2^2)^2}{64m_1^4m_2^4} - \frac{3(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2\mathbf{p}_1^2(\mathbf{p}_2^2)^2}{32m_1^4m_2^4} - \frac{7(\mathbf{p}_1^2)^2(\mathbf{p}_2^2)^2}{128m_1^4m_2^4}, \quad (\text{A4a})$$

$$H_{46}(\mathbf{x}_a, \mathbf{p}_a) = \frac{369(\mathbf{n}_{12} \cdot \mathbf{p}_1)^6}{160m_1^6} - \frac{889(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4\mathbf{p}_1^2}{192m_1^6} + \frac{49(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_1^2)^2}{16m_1^6} - \frac{63(\mathbf{p}_1^2)^3}{64m_1^6} - \frac{549(\mathbf{n}_{12} \cdot \mathbf{p}_1)^5(\mathbf{n}_{12} \cdot \mathbf{p}_2)}{128m_1^5m_2} \\ + \frac{67(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_1^2}{16m_1^5m_2} - \frac{167(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1^2)^2}{128m_1^5m_2} + \frac{1547(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4(\mathbf{p}_1 \cdot \mathbf{p}_2)}{256m_1^5m_2} - \frac{851(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{128m_1^5m_2} \\ + \frac{1099(\mathbf{p}_1^2)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{256m_1^3m_2} + \frac{3263(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2}{1280m_1^4m_2^2} + \frac{1067(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_1^2}{480m_1^4m_2^2} - \frac{4567(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2(\mathbf{p}_1^2)^2}{3840m_1^4m_2^2} \\ - \frac{3571(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1 \cdot \mathbf{p}_2)}{320m_1^4m_2^2} + \frac{3073(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{480m_1^4m_2^2} + \frac{4349(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{1280m_1^4m_2^2} \\ - \frac{3461\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{3840m_1^4m_2^2} + \frac{1673(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4\mathbf{p}_2^2}{1920m_1^4m_2^2} - \frac{1999(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2\mathbf{p}_1^2\mathbf{p}_2^2}{3840m_1^4m_2^2} + \frac{2081(\mathbf{p}_1^2)^2\mathbf{p}_2^2}{3840m_1^4m_2^2} - \frac{13(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)^3}{8m_1^3m_2^3} \\ + \frac{191(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)^3\mathbf{p}_1^2}{192m_1^3m_2^3} - \frac{19(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{384m_1^3m_2^3} - \frac{5(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{384m_1^3m_2^3} \\ + \frac{11(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{192m_1^3m_2^3} + \frac{77(\mathbf{p}_1 \cdot \mathbf{p}_2)^3}{96m_1^3m_2^3} + \frac{233(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_2^2}{96m_1^3m_2^3} - \frac{47(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_1^2\mathbf{p}_2^2}{32m_1^3m_2^3} \\ + \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)\mathbf{p}_2^2}{384m_1^3m_2^3} - \frac{185\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)\mathbf{p}_2^2}{384m_1^3m_2^3} - \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^4}{4m_1^2m_2^4} + \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_2)^4\mathbf{p}_1^2}{4m_1^2m_2^4} \\ - \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)^3(\mathbf{p}_1 \cdot \mathbf{p}_2)}{2m_1^2m_2^4} + \frac{21(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{16m_1^2m_2^4} + \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_2^2}{6m_1^2m_2^4} + \frac{49(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_1^2\mathbf{p}_2^2}{48m_1^2m_2^4} \\ - \frac{133(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1 \cdot \mathbf{p}_2)\mathbf{p}_2^2}{24m_1^2m_2^4} - \frac{77(\mathbf{p}_1 \cdot \mathbf{p}_2)^2\mathbf{p}_2^2}{96m_1^2m_2^4} + \frac{197(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_2^2)^2}{96m_1^2m_2^4} - \frac{173\mathbf{p}_1^2(\mathbf{p}_2^2)^2}{48m_1^2m_2^4} + \frac{13(\mathbf{p}_2^2)^3}{8m_1^2m_2^4}, \quad (\text{A4b})$$

$$H_{441}(\mathbf{x}_a, \mathbf{p}_a) = \frac{5027(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4}{384m_1^4} - \frac{22993(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2\mathbf{p}_1^2}{960m_1^4} - \frac{6695(\mathbf{p}_1^2)^2}{1152m_1^4} - \frac{3191(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)}{640m_1^3m_2} \\ + \frac{28561(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_1^2}{1920m_1^3m_2} + \frac{8777(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{384m_1^3m_2} + \frac{752969\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{28800m_1^3m_2} \\ - \frac{16481(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2}{960m_1^2m_2^2} + \frac{94433(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_1^2}{4800m_1^2m_2^2} - \frac{103957(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1 \cdot \mathbf{p}_2)}{2400m_1^2m_2^2} \\ + \frac{791(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{400m_1^2m_2^2} + \frac{26627(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2\mathbf{p}_2^2}{1600m_1^2m_2^2} - \frac{118261\mathbf{p}_1^2\mathbf{p}_2^2}{4800m_1^2m_2^2} + \frac{105(\mathbf{p}_2^2)^2}{32m_2^2}, \quad (\text{A4c})$$

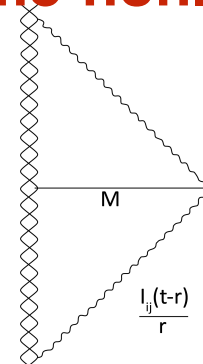
$$H_{442}(\mathbf{x}_a, \mathbf{p}_a) = \left(\frac{2749\pi^2}{8192} - \frac{211189}{19200} \right) \frac{(\mathbf{p}_1^2)^2}{m_1^4} + \left(\frac{63347}{1600} - \frac{1059\pi^2}{1024} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2\mathbf{p}_1^2}{m_1^4} + \left(\frac{375\pi^2}{8192} - \frac{23533}{1280} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^4}{m_1^4} \\ + \left(\frac{10631\pi^2}{8192} - \frac{1918349}{57600} \right) \frac{(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{m_1^2m_2^2} + \left(\frac{13723\pi^2}{16384} - \frac{2492417}{57600} \right) \frac{\mathbf{p}_1^2\mathbf{p}_2^2}{m_1^2m_2^2} \\ + \left(\frac{1411429}{19200} - \frac{1059\pi^2}{512} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2\mathbf{p}_1^2}{m_1^2m_2^2} + \left(\frac{248991}{6400} - \frac{6153\pi^2}{2048} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)(\mathbf{p}_1 \cdot \mathbf{p}_2)}{m_1^2m_2^2} \\ - \left(\frac{30383}{960} + \frac{36405\pi^2}{16384} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2}{m_1^2m_2^2} + \left(\frac{1243717}{14400} - \frac{40483\pi^2}{16384} \right) \frac{\mathbf{p}_1^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{m_1^3m_2} \\ + \left(\frac{2369}{60} + \frac{35655\pi^2}{16384} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^3(\mathbf{n}_{12} \cdot \mathbf{p}_2)}{m_1^3m_2} + \left(\frac{43101\pi^2}{16384} - \frac{391711}{6400} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)\mathbf{p}_1^2}{m_1^3m_2} \\ + \left(\frac{56955\pi^2}{16384} - \frac{1646983}{19200} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2(\mathbf{p}_1 \cdot \mathbf{p}_2)}{m_1^3m_2}, \quad (\text{A4d})$$

$$H_{421}(\mathbf{x}_a, \mathbf{p}_a) = \frac{64861\mathbf{p}_1^2}{4800m_1^2} - \frac{91(\mathbf{p}_1 \cdot \mathbf{p}_2)}{8m_1m_2} + \frac{105\mathbf{p}_2^2}{32m_2^2} - \frac{9841(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2}{1600m_1^2} - \frac{7(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)}{2m_1m_2}, \quad (\text{A4e})$$

$$H_{422}(\mathbf{x}_a, \mathbf{p}_a) = \left(\frac{1937033}{57600} - \frac{199177\pi^2}{49152} \right) \frac{\mathbf{p}_1^2}{m_1^2} + \left(\frac{176033\pi^2}{24576} - \frac{2864917}{57600} \right) \frac{(\mathbf{p}_1 \cdot \mathbf{p}_2)}{m_1m_2} + \left(\frac{282361}{19200} - \frac{21837\pi^2}{8192} \right) \frac{\mathbf{p}_2^2}{m_2^2} \\ + \left(\frac{698723}{19200} + \frac{21745\pi^2}{16384} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)^2}{m_1^2} + \left(\frac{63641\pi^2}{24576} - \frac{2712013}{19200} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_1)(\mathbf{n}_{12} \cdot \mathbf{p}_2)}{m_1m_2} \\ + \left(\frac{3200179}{57600} - \frac{28691\pi^2}{24576} \right) \frac{(\mathbf{n}_{12} \cdot \mathbf{p}_2)^2}{m_2^2}, \quad (\text{A4f})$$

$$H_{40}(\mathbf{x}_a, \mathbf{p}_a) = -\frac{m_1^4}{16} + \left(\frac{6237\pi^2}{1024} - \frac{169799}{2400} \right) m_1^3m_2 + \left(\frac{44825\pi^2}{6144} - \frac{609427}{7200} \right) m_1^2m_2^2. \quad (\text{A4g})$$

time-nonlocality due to tails G^4/c^8



$$H_{4\text{PN}}^{\text{nonloc}}(t) = -\frac{1}{5} \frac{G^2M}{c^8} I_{ij}^{(3)}(t) \\ \times \text{Pf}_{2r_{12}/c} \int_{-\infty}^{+\infty} \frac{dv}{|v|} I_{ij}^{(3)}(t+v),$$

Explicit 4PN EOB (non-spinning) dynamics (Damour-Jaranowski-Schaefer '14)

A **simple**, but crucial transformation between the real energy and the effective one:

$$\mathcal{E}_{\text{eff}} = \frac{(\mathcal{E}_{\text{real}})^2 - m_1^2 - m_2^2}{2(m_1 + m_2)},$$

A **simple(gauge-fixed) post-geodesic** effective

$$g_{\text{eff}}^{\mu\nu} P'_\mu P'_\nu + \mu^2 c^2 + Q(P'_\mu) = 0,$$

$$M = m_1 + m_2, \quad \mu = \frac{m_1 m_2}{m_1 + m_2}, \quad \nu = \frac{m_1 m_2}{(m_1 + m_2)^2}.$$

$$ds_{\text{eff}}^2 = -A(R; \nu) dt^2 + B(R; \nu) dR^2 + R^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

**Padé
resummed**

$$a_4 = \frac{94}{3} - \frac{41\pi^2}{32}; \quad a_5^c = \frac{2275\pi^2}{512} + \dots; \quad a'_5 = \dots$$

$$u \equiv \frac{GM}{R c^2}$$

$$A^{\text{PN}}(u; \nu) = 1 - 2u + 2\nu u^3 + \nu a_4 u^4 + (\nu a_5^c + \nu^2 a'_5 + \frac{64}{5} \nu \ln u) u^5$$

only

$$(AB)^{-1} = \bar{D}(u) = 1 + 6\nu u^2 + (52\nu - 6\nu^2) u^3 + \left(\left(-\frac{533}{45} - \frac{23761\pi^2}{1536} + \frac{1184}{15} \gamma_E - \frac{6496}{15} \ln 2 + \frac{2916}{5} \ln 3 \right) \nu \right. \\ \left. + \left(\frac{123\pi^2}{16} - 260 \right) \nu^2 + \frac{592}{15} \nu \ln u \right) u^4,$$

**gauge-invariant
information**

$$\hat{Q}(\mathbf{r}', \mathbf{p}') = \left(2(4 - 3\nu) \nu u^2 + \left(\left(-\frac{5308}{15} + \frac{496256}{45} \ln 2 - \frac{33048}{5} \ln 3 \right) \nu - 83\nu^2 + 10\nu^3 \right) u^3 \right) (\mathbf{n}' \cdot \mathbf{p}')^4 \\ + \left(\left(-\frac{827}{3} - \frac{2358912}{25} \ln 2 + \frac{1399437}{50} \ln 3 + \frac{390625}{18} \ln 5 \right) \nu - \frac{27}{5} \nu^2 + 6\nu^3 \right) u^2 (\mathbf{n}' \cdot \mathbf{p}')^6 + \mathcal{O}[\nu u (\mathbf{n}' \cdot \mathbf{p}')^8].$$

**small eccentricity
expansion of
H-nonlocal**

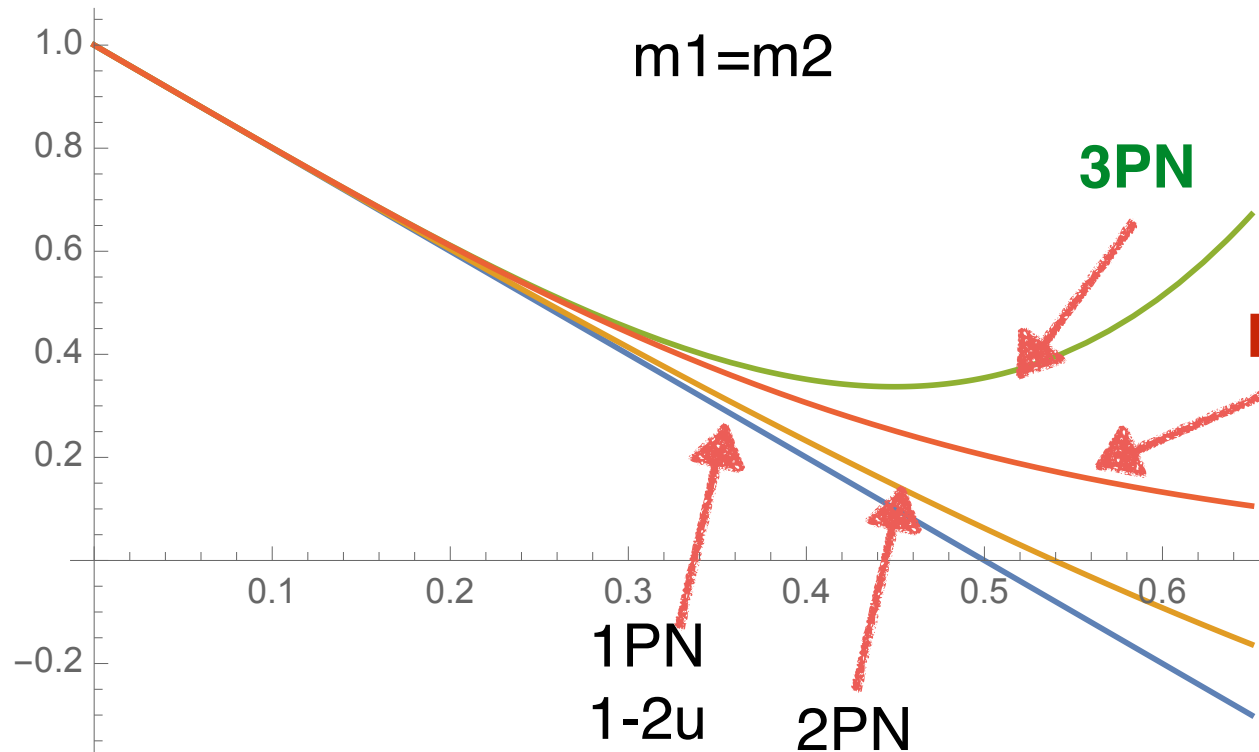
Resummed A(u) potential

$$ds_{\text{eff}}^2 = -A(R; \nu) dt^2 + B(R; \nu) dR^2 + R^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

$$u \equiv \frac{GM}{R c^2}$$

$$M = m_1 + m_2$$

$$\nu = \frac{m_1 m_2}{(m_1 + m_2)^2}$$



Spinning EOB effective Hamiltonian

Damour'01, Damour-Jaranowski-Schaefer'08, Barausse-Buonanno'11, Taracchini et al'12, Damour-Nagar'14,

$$H_{\text{eff}} = H_{\text{orb}} + H_{\text{so}} \quad \rightarrow \quad H_{\text{EOB}} = Mc^2 \sqrt{1 + 2\nu \left(\frac{H_{\text{eff}}}{\mu c^2} - 1 \right)}$$

$$\hat{H}_{\text{orb}}^{\text{eff}} = \sqrt{A \left(1 + B_p \mathbf{p}^2 + B_{np} (\mathbf{n} \cdot \mathbf{p})^2 - \frac{1}{1 + \frac{(\mathbf{n} \cdot \boldsymbol{\chi}_0)^2}{r^2}} \frac{(r^2 + 2r + (\mathbf{n} \cdot \boldsymbol{\chi}_0)^2)}{\mathcal{R}^4 + \Delta (\mathbf{n} \cdot \boldsymbol{\chi}_0)^2} ((\mathbf{n} \times \mathbf{p}) \cdot \boldsymbol{\chi}_0)^2 + Q_4 \right)}.$$

$$H_{\text{so}} = G_S \mathbf{L} \cdot \mathbf{S} + G_{S^*} \mathbf{L} \cdot \mathbf{S}^*,$$

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2; \quad \mathbf{S}_* = \frac{m_2}{m_1} \mathbf{S}_1 + \frac{m_1}{m_2} \mathbf{S}_2,$$

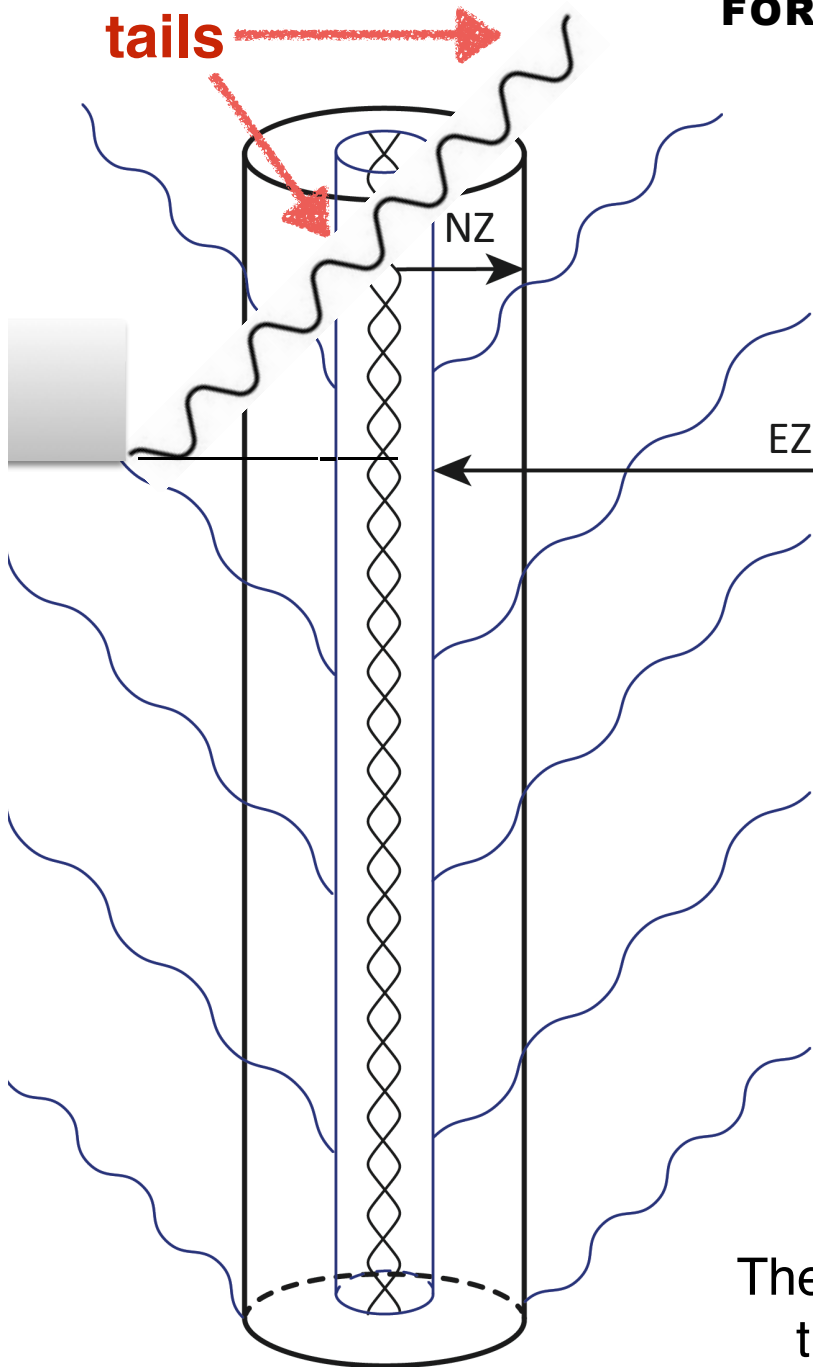
Gyrogravitomagnetic ratios (when neglecting spin² effects)

$$r^3 G_S^{\text{PN}} = 2 - \frac{5}{8} \nu u - \frac{27}{8} \nu p_r^2 + \nu \left(-\frac{51}{4} u^2 - \frac{21}{2} u p_r^2 + \frac{5}{8} p_r^4 \right) + \nu^2 \left(-\frac{1}{8} u^2 + \frac{23}{8} u p_r^2 + \frac{35}{8} p_r^4 \right)$$

$$r^3 G_{S_*}^{\text{PN}} = \frac{3}{2} - \frac{9}{8} u - \frac{15}{8} p_r^2 + \nu \left(-\frac{3}{4} u - \frac{9}{4} p_r^2 \right) - \frac{27}{16} u^2 + \frac{69}{16} u p_r^2 + \frac{35}{16} p_r^4 + \nu \left(-\frac{39}{4} u^2 - \frac{9}{4} u p_r^2 + \frac{5}{2} p_r^4 \right) \\ + \nu^2 \left(-\frac{3}{16} u^2 + \frac{57}{16} u p_r^2 + \frac{45}{16} p_r^4 \right)$$

GRAVITATIONAL WAVE GENERATION: MULTIPOLAR POST-MINKOWSKIAN

FORMALISM (BLANCHET-DAMOUR-IYER)



Decomposition of space-time in various overlapping regions:

1. **near-zone**: $r \ll \lambda$: PN
 2. **exterior zone**: $r \gg r_{\text{source}}$: MPM
 3. **far wave-zone**: Bondi-type expansion
- then **matching between the zones**

in exterior zone, **iterative solution** of Einstein's vacuum field equations by means of a **double expansion** in non-linearity and in multipoles, with crucial use of **analytic continuation** (complex B) for dealing with formal UV divergences at $r=0$

$$\begin{aligned}
 g &= \eta + Gh_1 + G^2h_2 + G^3h_3 + \dots, \\
 \square h_1 &= 0, \\
 \square h_2 &= \partial\partial h_1 h_1, \\
 \square h_3 &= \partial\partial h_1 h_1 h_1 + \partial\partial h_1 h_2, \\
 h_1 &= \sum_{\ell} \partial_{i_1 i_2 \dots i_{\ell}} \left(\frac{M_{i_1 i_2 \dots i_{\ell}}(t - r/c)}{r} \right) + \partial\partial \dots \partial \left(\frac{\epsilon_{j_1 j_2 k} S_{k j_3 \dots j_{\ell}}(t - r/c)}{r} \right), \\
 h_2 &= FP_B \square_{\text{ret}}^{-1} \left(\left(\frac{r}{r_0} \right)^B \partial\partial h_1 h_1 \right) + \dots, \\
 h_3 &= FP_B \square_{\text{ret}}^{-1} \dots
 \end{aligned}$$

STF tensors encoding multipole moments

mass-type and spin-type multipole moments

The PN-matched MPM formalism has allowed to compute the GW emission to very high accuracy (Blanchet et al)

Perturbative computation of GW flux from binary system

- lowest order : Einstein 1918 Peters-Mathews 63
- $1 + (v^2/c^2)$: Wagoner-Will 76
- ... + (v^3/c^3) : Blanchet-Damour 92, Wiseman 93
- ... + (v^4/c^4) : Blanchet-Damour-Iyer Will-Wiseman 95
- ... + (v^5/c^5) : Blanchet 96
- ... + (v^6/c^6) : Blanchet-Damour-Esposito-Farèse-Iyer 2004
- ... + (v^7/c^7) : Blanchet
- ... + $(v^8/c^8) + (v^9/c^9)$: Blanchet et al 2023-

$$\nu = \frac{m_1 m_2}{(m_1 + m_2)^2}$$

$$x = \left(\frac{v}{c}\right)^2 = \left(\frac{G(m_1 + m_2)\Omega}{c^3}\right)^{\frac{2}{3}} = \left(\frac{\pi G(m_1 + m_2)f}{c^3}\right)^{\frac{2}{3}}$$

using
MPM

LO
quadrupole
radiation

4PN

G⁴ beyond
LO (4-loop)

4.5PN

$$\begin{aligned} \mathcal{F} = \frac{32c^5}{5G} \nu^2 x^5 \Bigg\{ & 1 + \left(-\frac{1247}{336} - \frac{35}{12}\nu\right)x + 4\pi x^{3/2} + \left(-\frac{44711}{9072} + \frac{9271}{504}\nu + \frac{65}{18}\nu^2\right)x^2 + \left(-\frac{8191}{672} - \frac{583}{24}\nu\right)\pi x^{5/2} \\ & + \left[\frac{6643739519}{69854400} + \frac{16}{3}\pi^2 - \frac{1712}{105}\gamma_E - \frac{856}{105}\ln(16x) + \left(-\frac{134543}{7776} + \frac{41}{48}\pi^2\right)\nu - \frac{94403}{3024}\nu^2 - \frac{775}{324}\nu^3\right]x^3 \\ & + \left(-\frac{16285}{504} + \frac{214745}{1728}\nu + \frac{193385}{3024}\nu^2\right)\pi x^{7/2} \\ & + \left[-\frac{323105549467}{3178375200} + \frac{232597}{4410}\gamma_E - \frac{1369}{126}\pi^2 + \frac{39931}{294}\ln 2 - \frac{47385}{1568}\ln 3 + \frac{232597}{8820}\ln x \right. \\ & \quad \left. + \left(-\frac{1452202403629}{1466942400} + \frac{41478}{245}\gamma_E - \frac{267127}{4608}\pi^2 + \frac{479062}{2205}\ln 2 + \frac{47385}{392}\ln 3 + \frac{20739}{245}\ln x\right)\nu \right. \\ & \quad \left. + \left(\frac{1607125}{6804} - \frac{3157}{384}\pi^2\right)\nu^2 + \frac{6875}{504}\nu^3 + \frac{5}{6}\nu^4\right]x^4 \\ & + \left[\frac{265978667519}{745113600} - \frac{6848}{105}\gamma_E - \frac{3424}{105}\ln(16x) + \left(\frac{2062241}{22176} + \frac{41}{12}\pi^2\right)\nu \right. \\ & \quad \left. - \frac{133112905}{290304}\nu^2 - \frac{3719141}{38016}\nu^3\right]\pi x^{9/2} + \mathcal{O}(x^5) \Bigg\}. \end{aligned} \quad (4)$$

Resummed EOB waveform

Damour-Nagar 2007, Damour-Iyer-Nagar 2008

$$h_{\ell m} \equiv h_{\ell m}^{(N, \epsilon)} \hat{h}_{\ell m}^{(\epsilon)} \hat{h}_{\ell m}^{\text{NQC}}$$

$$\hat{h}_{\ell m}^{(\epsilon)} = \hat{S}_{\text{eff}}^{(\epsilon)} T_{\ell m} e^{i\delta_{\ell m}} \rho_{\ell m}^{\ell}$$

$$T_{\ell m} = \frac{\Gamma(\ell + 1 - 2i\hat{k})}{\Gamma(\ell + 1)} e^{\pi\hat{k}} e^{2i\hat{k} \ln(2kr_0)}$$

resums
an infinite
of leading
logs

$$\begin{aligned} \rho_{22}(x; \nu) = & 1 + \left(\frac{55\nu}{84} - \frac{43}{42} \right) x + \left(\frac{19583\nu^2}{42336} - \frac{33025\nu}{21168} - \frac{20555}{10584} \right) x^2 \\ & + \left(\frac{10620745\nu^3}{39118464} - \frac{6292061\nu^2}{3259872} + \frac{41\pi^2\nu}{192} - \frac{48993925\nu}{9779616} - \frac{428}{105} \text{eulerlog}_2(x) + \frac{1556919113}{122245200} \right) x^3 \\ & + \left(\frac{9202}{2205} \text{eulerlog}_2(x) - \frac{387216563023}{160190110080} \right) x^4 + \left(\frac{439877}{55566} \text{eulerlog}_2(x) - \frac{16094530514677}{533967033600} \right) x^5 + \mathcal{O}(x^6), \end{aligned}$$

$$\mathcal{F}_{\varphi} \equiv -\frac{1}{8\pi\Omega} \sum_{\ell=2}^{\ell_{\text{max}}} \sum_{m=1}^{\ell} (m\Omega)^2 |Rh_{\ell m}^{(\epsilon)}|^2$$

Recent developments: Cipriani+25, Ivanov+25 (a SWNS-revisit of MST)

EOB

$$\frac{dr}{dt} = \left(\frac{A}{B}\right)^{1/2} \frac{\partial \hat{H}_{\text{EOB}}}{\partial p_{r*}},$$

$$\frac{dp_{r*}}{dt} = - \left(\frac{A}{B}\right)^{1/2} \frac{\partial \hat{H}_{\text{EOB}}}{\partial r},$$

$$\Omega \equiv \frac{d\varphi}{dt} = \frac{\partial \hat{H}_{\text{EOB}}}{\partial p_\varphi},$$

$$\frac{dp_\varphi}{dt} = \hat{\mathcal{F}}_\varphi.$$

**Hamiltonian:
conservative
dynamics**

Rad Reac Force

**Resummed
waveform**

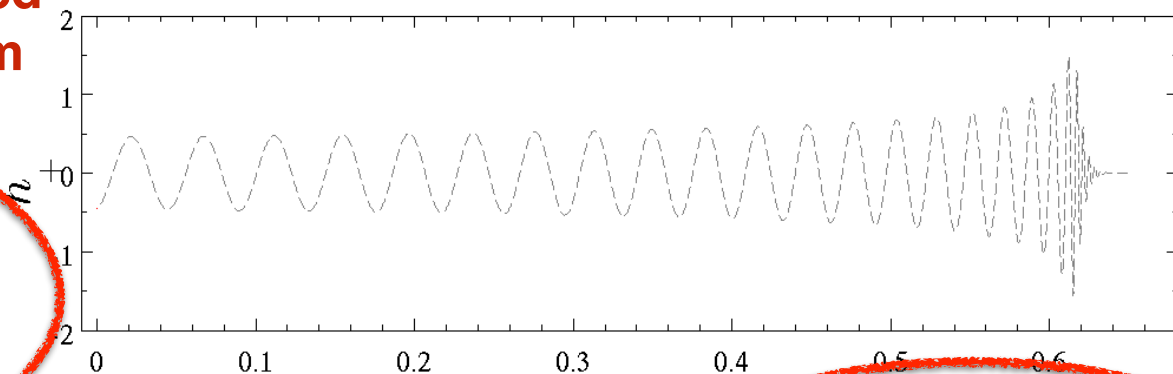
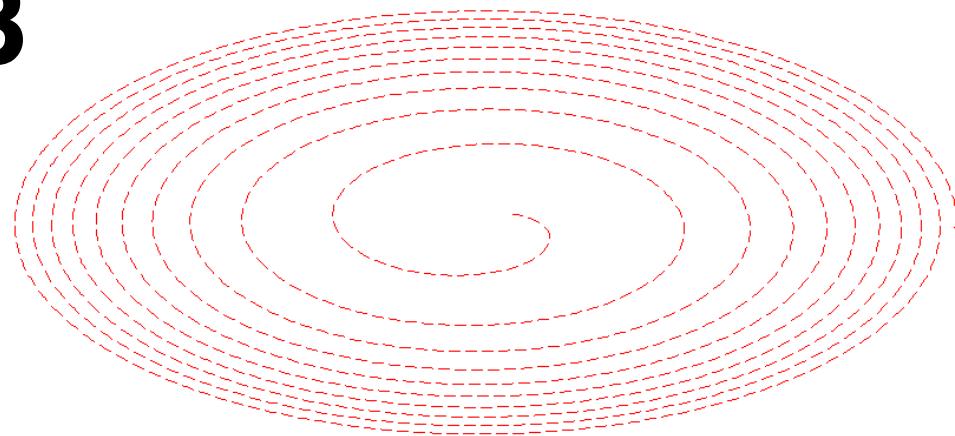
$$h_{\ell m} \equiv h_{\ell m}^{(N,\epsilon)} \hat{h}_{\ell m}^{(\epsilon)} \hat{h}_{\ell m}^{\text{NQC}}$$

$$\hat{h}_{\ell m}^{(\epsilon)} = \hat{S}_{\text{eff}}^{(\epsilon)} T_{\ell m} e^{i\delta_{\ell m}} \rho_{\ell m}^\ell$$

$$\mathcal{F}_\varphi \equiv -\frac{1}{8\pi\Omega} \sum_{\ell=2}^{\ell_{\text{max}}} \sum_{m=1}^{\ell} (m\Omega)^2 |Rh_{\ell m}^{(\epsilon)}|^2$$

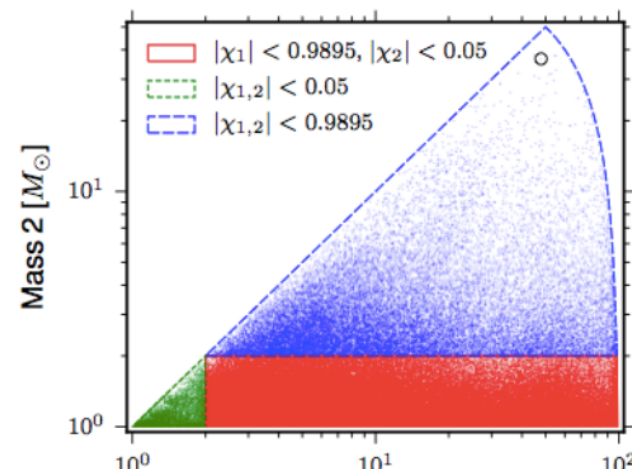
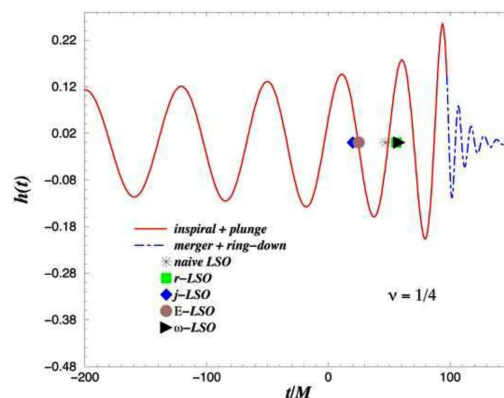
$$h_{\ell m}^{\text{ringdown}}(t) = \sum_N C_N e^{i\sigma_N(t-t_m)}$$

$$T_{\ell m} = \frac{\Gamma(\ell+1-2i\hat{k})}{\Gamma(\ell+1)} e^{\pi\hat{k}} e^{2i\hat{k}\log(2kr_0)},$$



$$h_{\ell m}^{\text{EOB}} = \theta(t_m - t) h_{\ell m}^{\text{insplunge}}(t) + \theta(t - t_m) h_{\ell m}^{\text{ringdown}}$$

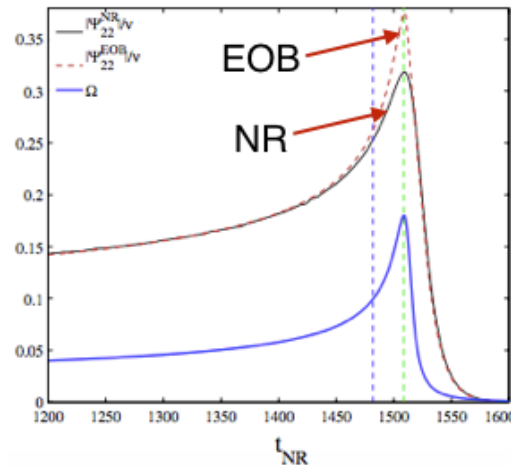
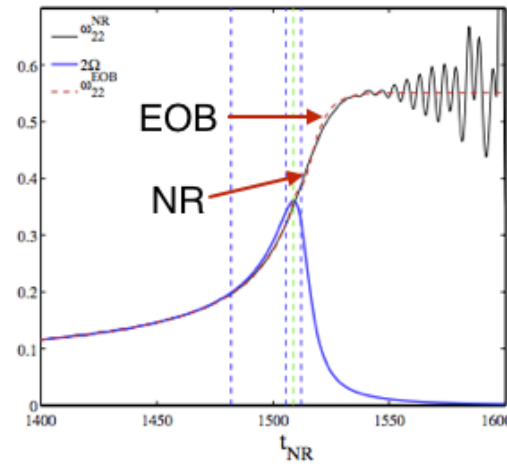
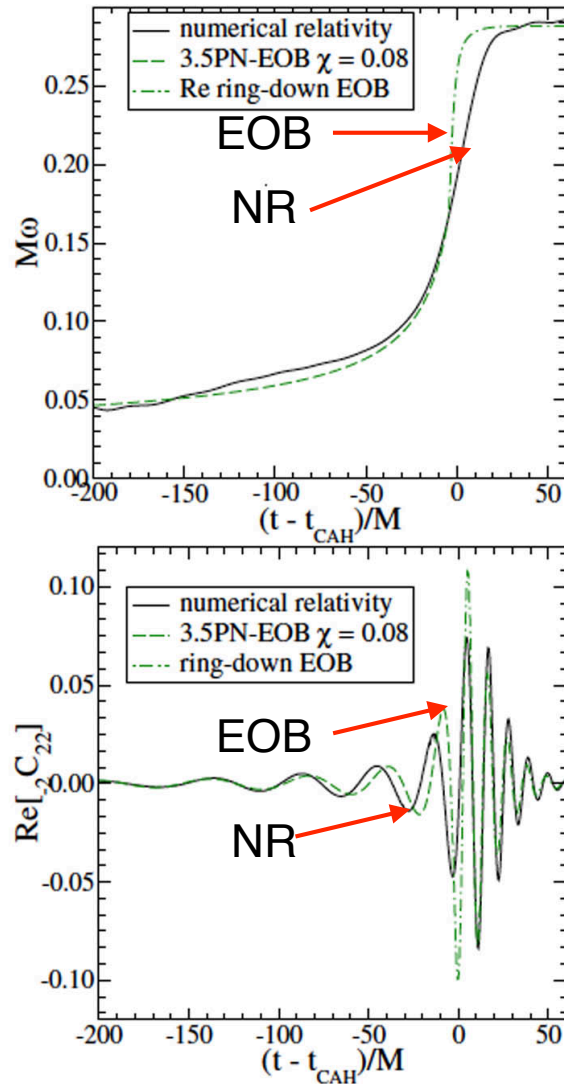
**Complete waveforms
for BBH coalescences:**



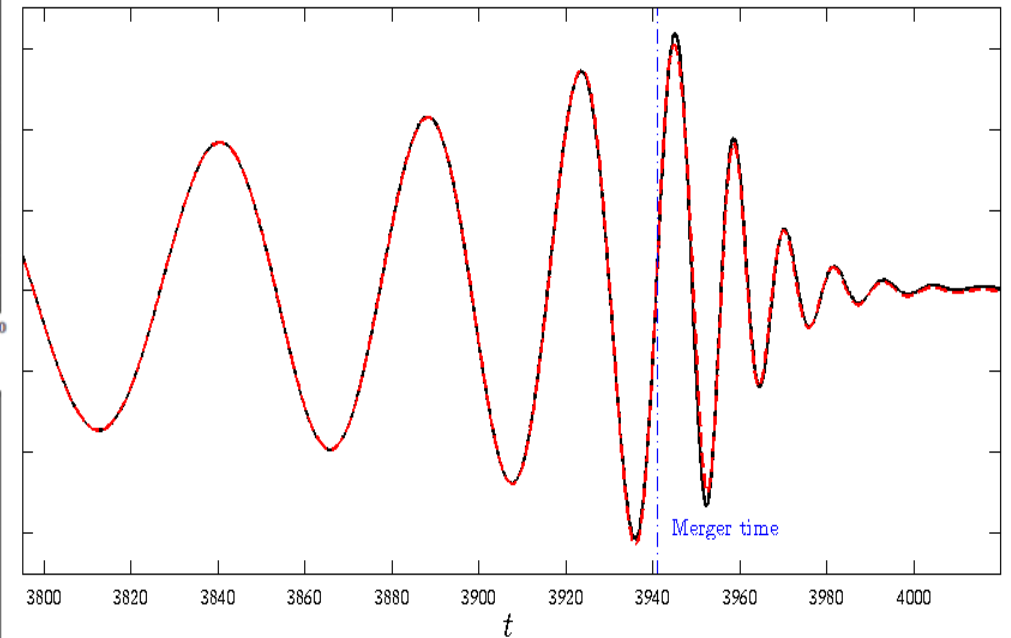
From EOB vs NR to EOB-NR waveforms

Buonanno-Cook-Pretorius 2007

TD-Nagar-Dorband-
Pollney-Rezzolla 2008



EOB-NR vs NR



**EOB-NR is obtained by
tuning some yet unknown
theoretical EOB parameters
to a sample of NR simulations,
e.g. $+ a_6^{\text{c(nu)}} u^6$**

FIG. 21 (color online). We compare the NR and EOB frequency and $\text{Re}[C_{22}]$ waveforms throughout the entire inspiral–merger–ring-down evolution. The data refers to the $d = 16$ run.

NR-completed resummed 5PN EOB radial A potential

« We think, however, that a suitable “numerically fitted” and, if possible, “analytically extended” EOB Hamiltonian should be able to fit the needs of upcoming GW detectors. » (TD 2001)

here Damour-Nagar-Bernuzzi '13, Nagar-et al '16; alternative: Taracchini et al '14, Bohe et al '17

4PN analytically complete + 5 PN logarithmic term in the $A(u, \nu)$ function [$u = GM/R$, $\nu = m_1 m_2 / (m_1 + m_2)^2$]
[Damour 09, Blanchet et al 10, Barack-Damour-Sago 10, Le Tiec et al 11, Barausse et al 11, Akcay et al 12, Bini-Damour 13, Damour-Jaranowski-Schäfer 14, Nagar-Damour-Reisswig-Pollney 15]

$$A(u; \nu, a_6^c) = P_5^1 \left[1 - 2u + 2\nu u^3 + \nu \left(\frac{94}{3} - \frac{41}{32} \pi^2 \right) u^4 \right.$$

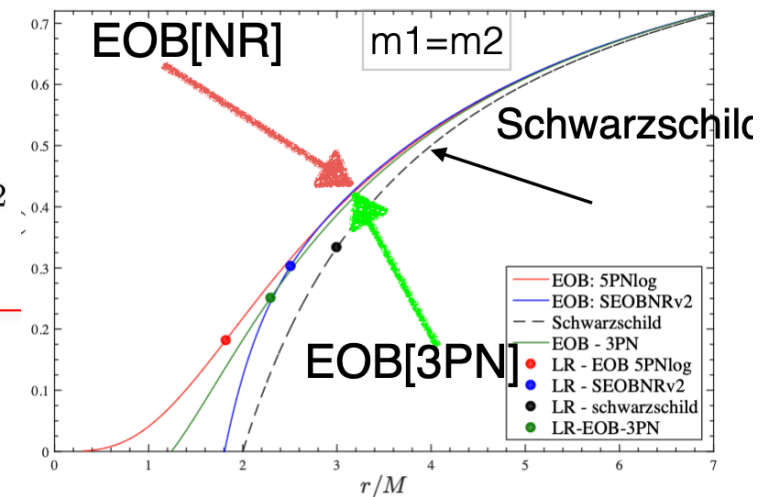
$$+ \nu \left[-\frac{4237}{60} + \frac{2275}{512} \pi^2 + \left(-\frac{221}{6} + \frac{41}{32} \pi^2 \right) \nu + \frac{64}{5} \ln(16e^{2\gamma} u) \right] u^5$$

$$+ \nu \left[a_6^c(\nu) - \left(\frac{7004}{105} + \frac{144}{5} \nu \right) \ln u \right] u^6 \Big]$$

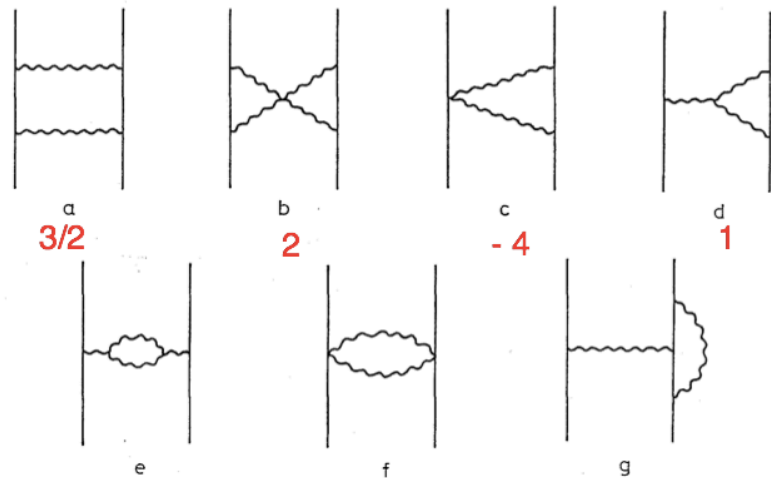
$$u = \frac{GM}{c^2 R}$$

$$\nu = \frac{m_1 m_2}{(m_1 + m_2)^2}$$

$$a_6^{c \text{ NR-tuned}}(\nu) = 81.38 - 1330.6 \nu + 3097.3 \nu^2$$



Quantum Scattering Amplitudes and 2-body Dynamics



Quantum Scattering Amplitudes → Potential

one-graviton exchange :

Corinaldesi '56 '71,

Barker-Gupta-Haracz 66,

Barker-O'Connell 70, Hiida-Okamura72

Nonlinear: Iwasaki 71 [1PN],

Okamura-Ohta-Kimura-Hiida 73[2 PN]

Using modern amplitude techniques: Bjerrum-Bohr+'03

Amati-Ciafaloni-Veneziano 1987-2008

Ultra-High-Energy ($s \gg M_{\text{Planck}}^2$)

Four-graviton Scattering at 2 loops

Eikonal phase δ in $D=4$

with one- and two-loop corrections using the Regge-Gribov approach

Fig. 3. The "H" diagram that provides the leading correction to the eikonal.

$$\delta = \frac{Gs}{\hbar} \left(\log \left(\frac{L_{IR}}{b} \right) + \frac{6\ell_s^2}{\pi b^2} + \frac{2G^2 s}{b^2} \left(1 + \frac{2i}{\pi} \log(\cdots) \right) \right)$$

confirmed by DiVecchia+'19

Personally becoming aware of the ACV results in Parma 2008, plus discussions at IHES with Donoghue and Vanhove → GSF and EOB (TD 2010): scattering and zero-binding zoom-whirl orbit (Barack et al'19)

Having so computed $\mathcal{E}$ and J one might then, for instance, compare the EOB prediction for the scattering angle $\theta(\mathcal{E}, J)$ (which follows from the EOB Hamiltonian) with GSF computations of θ for a sample of values of $\mathcal{E}$ and J . We see that, in principle, we have access here to one function of *two* real variables, which is ample information for determining the functions entering the EOB formalism.

PM approach to Two-Body Dynamics

Bertotti'56, Havas-Goldberg'62,

Rosenblum'78, Westpfahl'79, Portilla'80, Bel et al.81

PM revival via two-body scattering

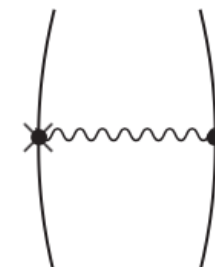
Gravitational scattering, post-Minkowskian approximation,
and effective-one-body theory

TD 2016, 2017: High-energy gravitational scattering and the general relativistic two-body problem G^1

A technique for translating the classical scattering function of two gravitationally interacting bodies into a corresponding (effective one-body) Hamiltonian description has been recently introduced [Phys. Rev. D **94**, 104015 (2016)]. Using this technique, we derive, for the first time, to second-order in Newton's constant (i.e. one classical loop) the Hamiltonian of two point masses having an arbitrary (possibly relativistic) relative velocity. The resulting (second post-Minkowskian) Hamiltonian is found to have a tame high-energy structure which we relate both to gravitational self-force studies of large mass-ratio binary systems, and to the ultra high-energy quantum scattering results of Amati, Ciafaloni and Veneziano. We derive several consequences of our second post-Minkowskian Hamiltonian: (i) the need to use special phase-space gauges to get a tame high-energy limit; and (ii) predictions about a (rest-mass independent) linear Regge trajectory behavior of high-angular-momenta, high-energy circular orbits. Ways of testing these predictions by dedicated numerical simulations are indicated. We finally indicate a way to connect our classical results to the quantum gravitational scattering amplitude of two particles, and we urge amplitude experts to use their novel techniques to compute the two-loop scattering amplitude of scalar masses, from which one could deduce the third post-Minkowskian effective one-body Hamiltonian.

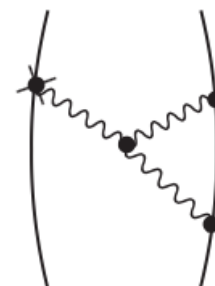
tree-level

G^1



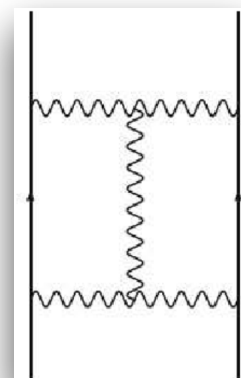
one-loop

G^2



two-loop

G^3+G^4



Cheung-Rothstein-Solon 2018

From Scattering Amplitudes to Classical Potentials in the Post-Minkowskian Expansion

We combine tools from effective field theory and generalized unitarity to construct a map between on-shell scattering amplitudes and the classical potential for interacting spinless particles. For general relativity, we obtain analytic expressions for the classical potential of a binary black hole system at second order in the gravitational constant and all orders in velocity. Our results exactly match all known results up to fourth post-Newtonian order, and offer a simple check of future higher order calculations. By design, these methods should extend to higher orders in perturbation theory.

one-loop

G^2

PM scattering results (here without spin)

3PM=G³ ultrarelativistic $\gamma \rightarrow \infty$ Amati-Ciafaloni-Veneziano'90
Bern-Cheung-Roiban-Shen-Solon-Zeng'19

inclusion of radiative effects TD'20, DiVecchia+'21, Hermann+21, ...

J loss at G² (TD'20): **subtleties linked to zero-freq gravitons and BMS** (Ven-Vilk22)

$$\chi^{\text{rr}} = -\frac{1}{2} \frac{\partial \chi^{\text{cons}}(E, J)}{\partial J} J^{\text{rad}} - \frac{1}{2} \frac{\partial \chi^{\text{cons}}(E, J)}{\partial E} E^{\text{rad}}$$

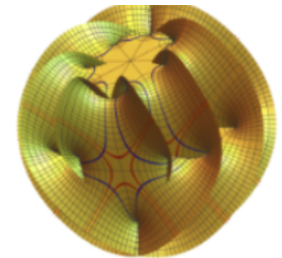
conservative: Bern+'22, Dlapa+'22

4PM=G⁴ including radiation-reaction: Dlapa+'23, Damgaard+'23

$$\mathcal{F}_{\text{rad-reac}} = O(G^2/c^5) \implies \exists \mathcal{F}_{\text{rad-reac}}^2 = O(G^4) \text{ effects} \quad (\text{Bini-TD-Geralico'21})$$

5PM=G⁵ **Driesse+'24'26, Bern+'25'26**

CY3



$$(\Delta p_a^\mu)^{5\text{PM}} \sim \frac{G^5}{b^5} m_1 m_2 (m_2^4 + m_1 m_2^3 + m_1^2 m_2^2 + m_1^3 m_2 + m_1^4)$$

probe
1SF
2SF ?
1SF
probe

Two equivalent **gauge-invariant** routes to derive the EOB dynamics

Delaunay Hamiltonian BuonannoTD

Scattering angle TD'16-18

$$E = H_D(I_r, I_\theta, I_\phi)$$

'99

$$\frac{1}{2}\chi = \Phi(E_{\text{real}}, J; m_1, m_2, G)$$

$$0 = g_{\text{eff}}^{\mu\nu} P_\mu P_\nu + \mu^2 + Q$$

$$\mathcal{E}_{\text{eff}} = \frac{(\mathcal{E}_{\text{real}})^2 - m_1^2 - m_2^2}{2(m_1 + m_2)}$$

$$\frac{\mathcal{E}_{\text{eff}}}{\mu} = \gamma = -\frac{p_1 \cdot p_2}{m_1 m_2}$$

$$\chi(\gamma, j) = 2\frac{\chi_1(\gamma)}{j} + 2\frac{\chi_2(\gamma)}{j^2} + 2\frac{\chi_3(\gamma)}{j^3} + 2\frac{\chi_4(\gamma)}{j^4} + O\left[\frac{1}{j^5}\right]$$

$$\frac{1}{j} = \frac{Gm_1 m_2}{J}$$

$g_{\text{eff}}^{\mu\nu}$ = Schwarzschild metric $M=m_1+m_2$

$$Q = \left(\frac{GM}{R}\right)^2 q_2(E) + \left(\frac{GM}{R}\right)^3 q_3(E) + O(G^4)$$

$$q_2 = -\frac{4}{\pi}(\chi_2 - \chi_2^{\text{Schw}})$$

$$q_3 = \frac{4}{\pi} \frac{2\gamma^2 - 1}{\gamma^2 - 1} (\chi_2 - \chi_2^{\text{Schw}}) - \frac{\chi_3 - \chi_3^{\text{Schw}}}{\gamma^2 - 1}$$

Various ways of formulating the EOB mass-shell condition

$$g^{\mu\nu}(X^\lambda)P_\mu P_\nu + \mu^2 + Q(X^\mu, P_\mu) = 0$$

-P₀=gamma=H_{eff}

expressed as a function
of R and P_R² or **P**²

Traditionally DJS gauge: $Q(R, P_R) \sim (GM/R)^2 P_r^4 + \dots$
convenient both for solving $E=H(X, P)$ and for mildly-eccentric dynamics

in PM gravity: TD 1710.10599 introduced various gauges:

post-Schwarzschild: $g_{\text{eff}}^{\mu\nu}$ = Schwarzschild metric $M=m_1+m_2$ $Q = \left(\frac{GM}{R}\right)^2 q_2(E) + \left(\frac{GM}{R}\right)^3 q_3(E) + O(G^4)$

absorbing Q in A(g,R): $A_{S^*}(R, \gamma, \nu) = \frac{A_S(R)}{1 - \frac{1}{\gamma^2} A_S(R) Q(R, \gamma, \nu)}$

Newtonianlike
potential:

$$p_{\bar{r}}^2 + \frac{j^2}{\bar{r}^2} = p_\infty^2 + w(\bar{r}, \gamma).$$

$$w(\bar{r}, \gamma) = \frac{w_1(\gamma)}{\bar{r}} + \frac{w_2(\gamma)}{\bar{r}^2} + \frac{w_3(\gamma)}{\bar{r}^3} + \frac{w_4(\gamma)}{\bar{r}^4} + O\left[\frac{1}{\bar{r}^5}\right]$$

**New issues
posed by PM**

? solving the mass-shell condition to get $E=H(X, P)$
? choosing a « gauge » for the effective metric
? choosing a resummation

A novel Lagrange-multiplier approach to the effective-one-body dynamics of binary systems in post-Minkowskian gravity

Damour-Nagar-Placidi-Rettegno, March 2025

a geodesic-like mass-shell condition involving an energy-dependent effective metric:

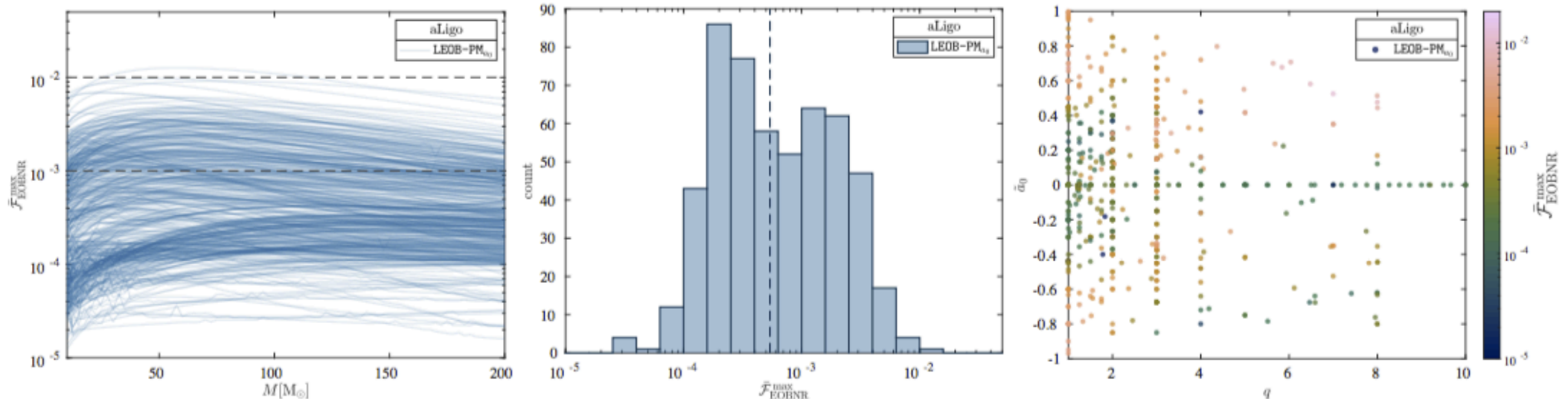
$$g_*^{\mu\nu}(X^\mu, \gamma) P_\mu P_\nu + \mu^2 = 0$$

up to now solved perturbatively
wrt the energy $\gamma = -P_0/\mu$;
this led to several drawbacks

Lagrange multiplier approach:

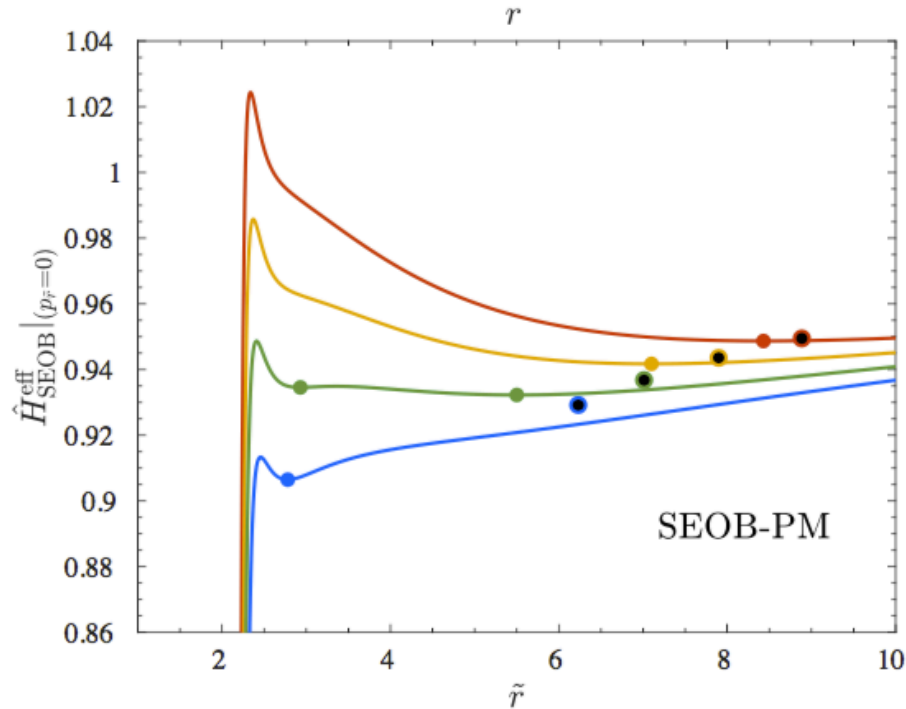
$$\mathcal{C} = g_*^{\mu\nu}(x^\lambda, p_0) p_\mu p_\nu + 1$$

$$S[x^\mu, p_\mu, e_L] = \int \left[p_\mu \frac{dx^\mu}{d\tau} - e_L \mathcal{C}(x^\mu, p_\mu) \right] d\tau$$

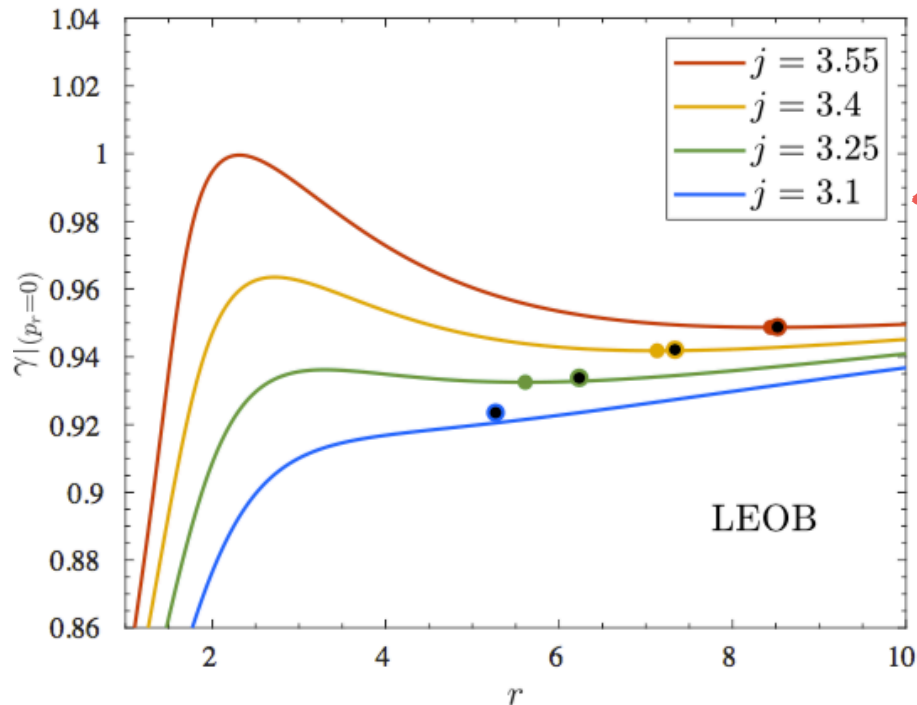


median unfaithfulness= 5.39×10^{-4} ;

Radial potentials for inspiralling orbits in two versions of EOB-PM



when iteratively
solving the
mass-shell condition
to get $H_{\text{eff}}(R, J)$
SEOB-PM, Buonanno+24



when exactly
solving the
mass-shell condition
to get $H_{\text{eff}}(R, J)$
LEOB, Damour+25

Mass polynomiality structure in scattering (TD'20)

$$\frac{dx_a^\mu}{ds_a} = g^{\mu\nu}(x_a) u_{a\nu},$$

$$R^{\mu\nu} - \frac{1}{2} R g^{\mu\nu} = 8\pi G T^{\mu\nu},$$

$$\frac{du_{a\mu}}{ds_a} = -\frac{1}{2} \partial_\mu g^{\alpha\beta}(x_a) u_{a\alpha} u_{a\beta},$$

$$T^{\mu\nu}(x) = \sum_{a=1,2} m_a \int ds_a u_a^\mu u_a^\nu \frac{\delta^4(x - x_a(s_a))}{\sqrt{g}}$$

$$\Delta p_{a\mu} = -\frac{m_a}{2} \int_{-\infty}^{+\infty} ds_a \partial_\mu g^{\alpha\beta}(x_a) u_{a\alpha} u_{a\beta}$$

$$\Delta p_{1\mu} = -2Gm_1m_2 \frac{2(u_{10} \cdot u_{20})^2 - 1}{\sqrt{(u_{10} \cdot u_{20})^2 - 1}} \frac{b_\mu}{b^2} + \frac{Gm_1m_2}{b} \Delta_\mu.$$

polynomial in Gm_1/b and Gm_2/b

conservative scattering case

$$\frac{1}{2} \chi(E_{\text{real}}, J) = \frac{\chi_1(\gamma, \nu)}{j} + \frac{\chi_2(\gamma, \nu)}{j^2} + \frac{\chi_3(\gamma, \nu)}{j^3} + \frac{\chi_4(\gamma, \nu)}{j^4} + \dots,$$

at G^n

$$h^{n-1}(\gamma, \nu) \chi_n(\gamma, \nu) = P_{d(n)}^\gamma(\nu),$$

polynomial in ν of degree $[(n-1)/2]$

0SF scattering gives access to full G^2 dynamics !

1SF scattering gives access to full G^3 and G^4 conservative dynamics !

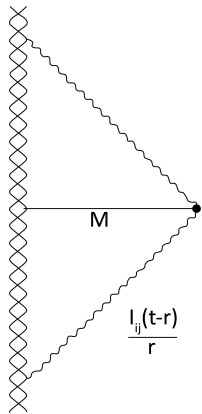
2SF scattering gives access to full G^5 and G^6 conservative dynamics



Tutti-Frutti method (Bini-TD-Geralico,'19,20,21)

with PN local Hamiltonian

$$S_{\text{tot}}^{\leq n\text{PN}}[x_1(s_1), x_2(s_2)] = S_{\text{loc}}^{\leq n\text{PN}}[x_1(s_1), x_2(s_2)] + S_{\text{nonloc}}^{\leq n\text{PN}}[x_1(s_1), x_2(s_2)].$$



$$S_{\text{nonloc}}^{4+5\text{PN}}[x_1(s_1), x_2(s_2)] = \frac{G^2 \mathcal{M}}{c^3} \int dt \text{PF}_{2r_{12}^h(t)/c} \mathcal{F}_{\text{IPN}}^{\text{split}}(t, t') \times \int \frac{dt'}{|t-t'|} \mathcal{F}_{\text{IPN}}^{\text{split}}(t, t'),$$

$$\mathcal{F}_{\text{IPN}}^{\text{split}}(t, t') = \frac{G}{c^5} \left(\frac{1}{5} I_{ab}^{(3)}(t) I_{ab}^{(3)}(t') + \frac{1}{189c^2} I_{abc}^{(4)}(t) I_{abc}^{(4)}(t') + \frac{16}{45c^2} J_{ab}^{(3)}(t) J_{ab}^{(3)}(t') \right). \quad (1)$$

starting at 5.5PN, G^5

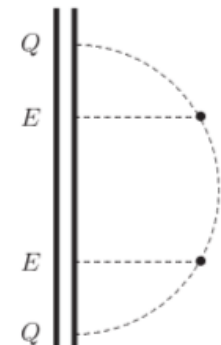
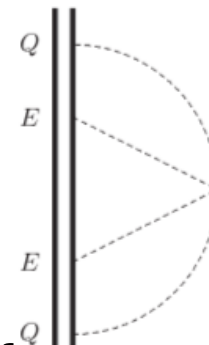
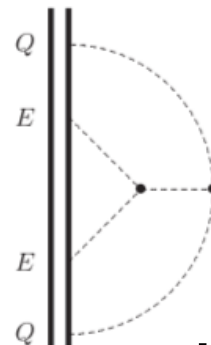
Combining:

1GSF Mano-Suzuki-Takasugi-based computations of

Detweiler redshift+ first law (LeTiec-Blanchet-Whiting)

polynomiality in ν

Delaunay averaging, one determines H_{loc} modulo a few 2SF parameters



(b)

(c)

Tutti-Frutti method

SIXTH POST-NEWTONIAN LOCAL-IN-TIME DYNAMIC

6PN

**conservative
dynamics
complete at
3PM and 4PM**



(Bini-TD-Geralico
'19,'20'21)

**combines
PN, MPM, EOB,
Delaunay,
Self-Force,
mass-
polynomiality
of scattering
angle**

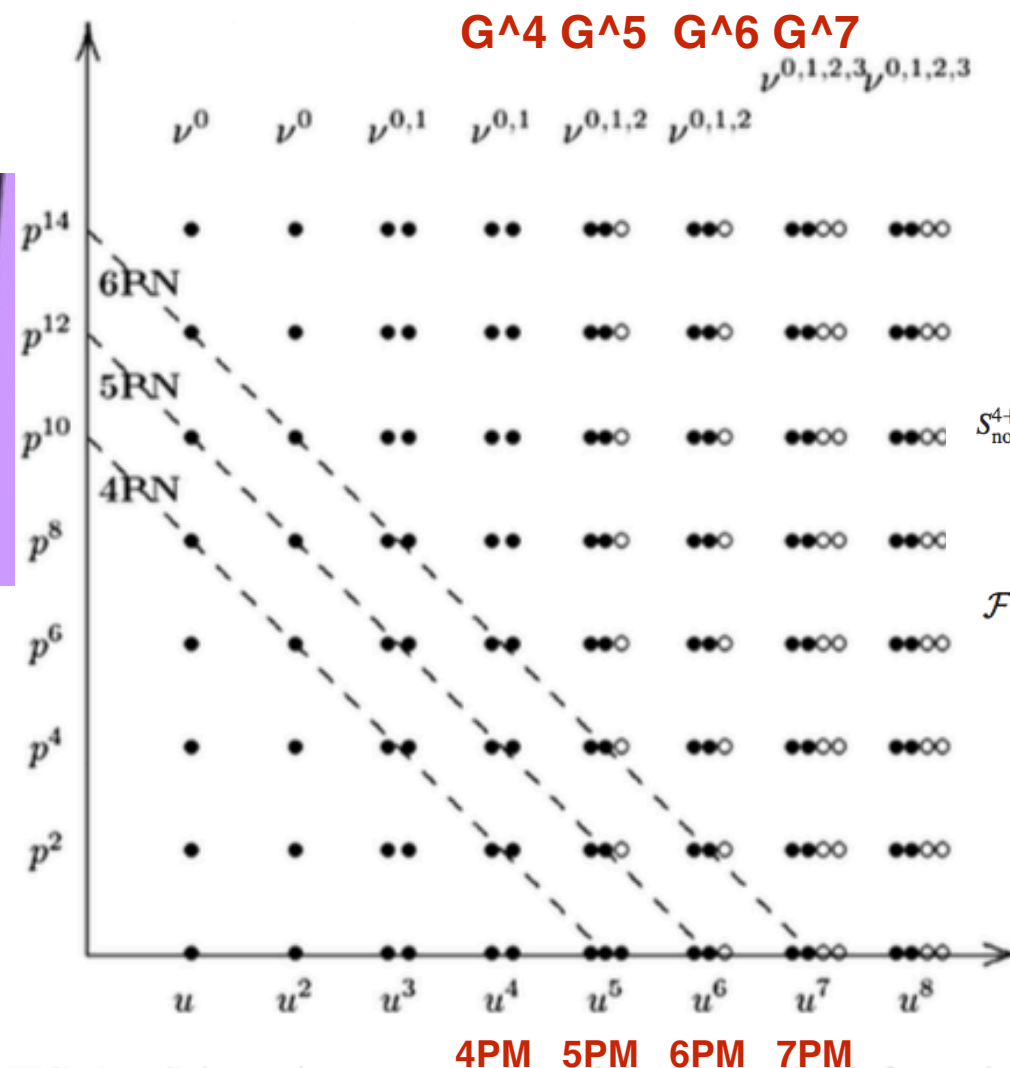


FIG. 1. Schematic representation of the irreducible information contained, at each post-Minkowskian level (keyed by a power of $u = GM/r$), in the local dynamics. Each vertical column of dots describes the post-Newtonian expansion (keyed by powers of p^2) of an energy-dependent function parametrizing the scattering angle. The various columns at a given post-Minkowskian level correspond to increasing powers of the symmetric mass-ratio ν . See text for details.

$$S_{\text{nonloc}}^{4+5\text{PN}}[x_1(s_1), x_2(s_2)] = \frac{G^2 \mathcal{M}}{c^3} \int dt \text{PF}_{2r_{12}^h(t)/c} \\ \times \int \frac{dt'}{|t-t'|} \mathcal{F}_{1\text{PN}}^{\text{split}}(t, t').$$

$$\mathcal{F}_{1\text{PN}}^{\text{split}}(t, t') = \frac{G}{c^5} \left(\frac{1}{5} I_{ab}^{(3)}(t) I_{ab}^{(3)}(t') + \frac{1}{189 c^2} \right. \\ \left. + \frac{16}{45 c^2} J_{ab}^{(3)}(t) J_{ab}^{(3)}(t') \right).$$

5PN

**conservative
complete at
5PM and 6PM
modulo
d5 and a6**

**compar. EFT
Bluemlein+
Porto-Riva'25,26**

Tutti Frutti vs Worldline Effective Field Theory (Bini-TD'25)

$$\Delta p_a^\mu = \Delta p_a^{\text{cons} \mu} + \Delta p_a^{\text{rrlin} J_{\text{rad}} \mu} + \Delta p_a^{\text{rrlin} P_{\text{rad}} \mu} + \Delta p_1^{\text{rr remain sup} \mu}$$

linear-response formula
Bini-TD, Bini-TD-Geralico

at G^4

$$c_{1b,1\text{rad}}^{4\text{diss}} = \frac{G^4}{b^4} m_1^2 m_2^2 \left\{ (m_1 + m_2) \left[\mathcal{E}(\gamma) \frac{\gamma(6\gamma^2 - 5)}{(\gamma^2 - 1)^{3/2}} - \pi \frac{3}{4} \hat{J}_2(\gamma) \frac{(5\gamma^2 - 1)}{(\gamma^2 - 1)^{3/2}} - \hat{J}_3(\gamma) \frac{(2\gamma^2 - 1)}{(\gamma^2 - 1)^2} \right] - m_1 \mathcal{E}(\gamma) \frac{2\gamma^2 - 1}{(\gamma + 1)\sqrt{\gamma^2 - 1}} \right\}. \quad (3.2)$$

uniquely
determined at G^4
by mass polynomiality
and rad-reac structure

determines J_3 from $cb1\text{rad}$

$$c_{1b,2\text{rad}}^{4\text{diss}} = \frac{m_1}{m_2 - m_1} P_{xG^4}^{\text{rad}}$$

direct link
between
 $cb2\text{rad}$ and P_x^{rad}

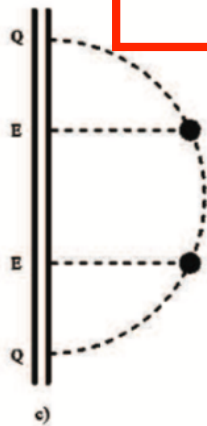
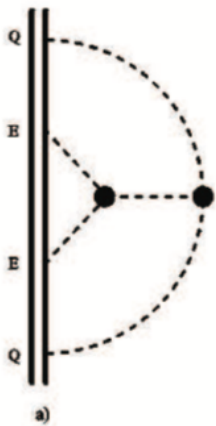
at G^5 1SF

$$f_b^{G^5, 1\text{SF}} = -\frac{2\tilde{\chi}_5^{1\text{cons}}}{(\gamma^2 - 1)^2} - \frac{\chi_1 \hat{J}_4^0}{b^5 (\gamma^2 - 1)^2} + K$$

TF determined to 6PN
including a **tail-of-tail** term

determined to 6PN
by TF + Heissenberg'24

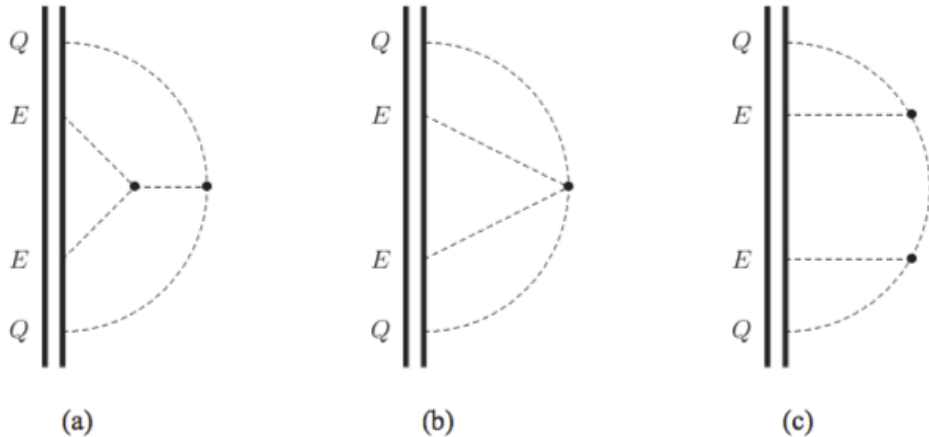
TF determined P_μ^{rad} to 5.5 PN



High-post-Newtonian-order dynamical effects induced by tail-of-tail interactions in a two body system (BDG'25)

tail-of-tail conservative action

(DJS'14,BD'25,using Blanchet'05,Goldberger-Ross'10,...)



$$S_{\text{tail-of-tail time-sym}}^{\text{tail-of-tail}} = \frac{1}{2} \left(\frac{GM}{c^3} \right)^2 G \sum_{l \geq 2} \frac{1}{c^{2l+1}} a_l \beta_l^{\text{even}} \int dt \times$$

$$\int_{-\infty}^{\infty} dt' I_L^{(l+2)}(t) I_L^{(l+2)}(t') \ln \frac{c|t-t'|}{2r_0}$$

$$+ \frac{1}{2} \left(\frac{GM}{c^3} \right)^2 G \sum_{l \geq 2} \frac{1}{c^{2l+3}} b_l \beta_l^{\text{odd}} \int dt \times$$

$$\int_{-\infty}^{\infty} dt' J_L^{(l+2)}(t) J_L^{(l+2)}(t') \ln \frac{c|t-t'|}{2r_0},$$

1SF confirmations and 2SF new results at the 6.5PN level

$$A(u, \nu) = 1 - 2u + \sum_{n \geq 3} a_n(\nu, \ln u) u^n, \quad a_{6.5} = \frac{13696}{525} \nu \pi,$$

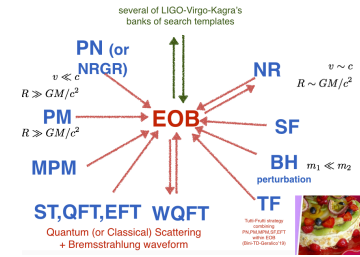
$$\bar{D}(u, \nu) = 1 + \sum_{n \geq 2} \bar{d}_n(\nu, \ln u) u^n, \quad a_{7.5} = -\frac{10052}{225} \nu^2 \pi - \frac{512501}{3675} \nu \pi,$$

agreement at 6.5PN with 1SF Driesse et al (using a recent result of Geralico'25)

prediction of the « conservative » G^5 scattering at 2SF in terms of TF undetermined parameters

Strong-field scattering of two black holes: Numerical relativity meets post-Minkowskian gravity

Thibault Damour¹ and Piero Retteno^{2,3} (PRD March 2023)



$w(r)$ introduced
in TD 1710.10599

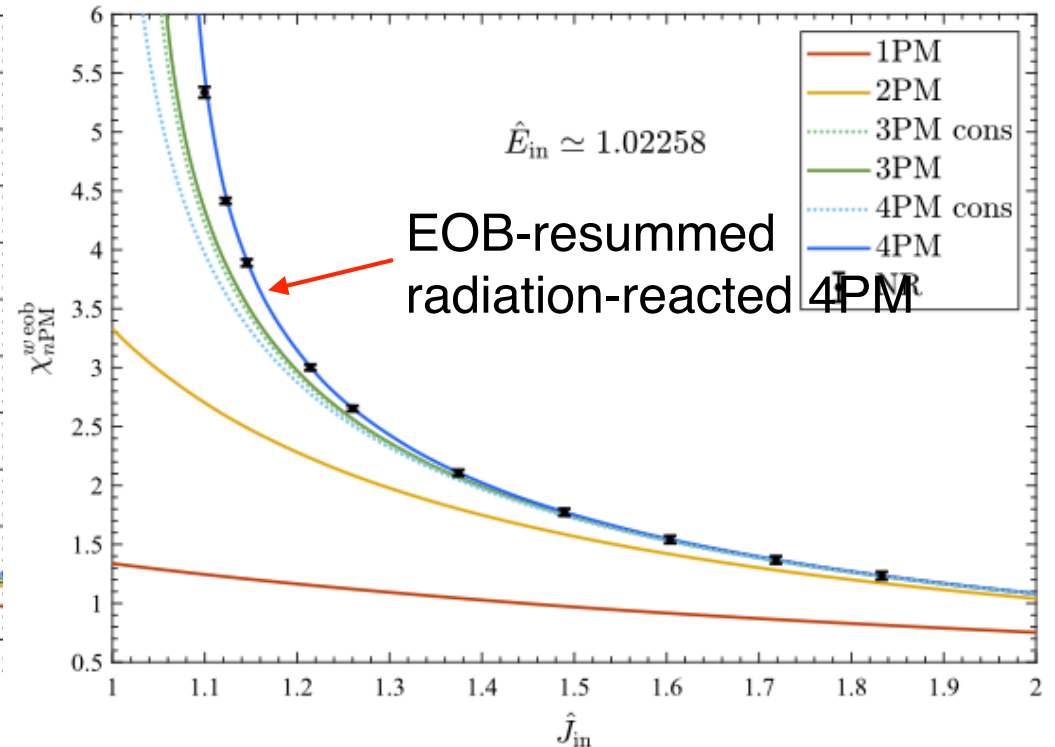
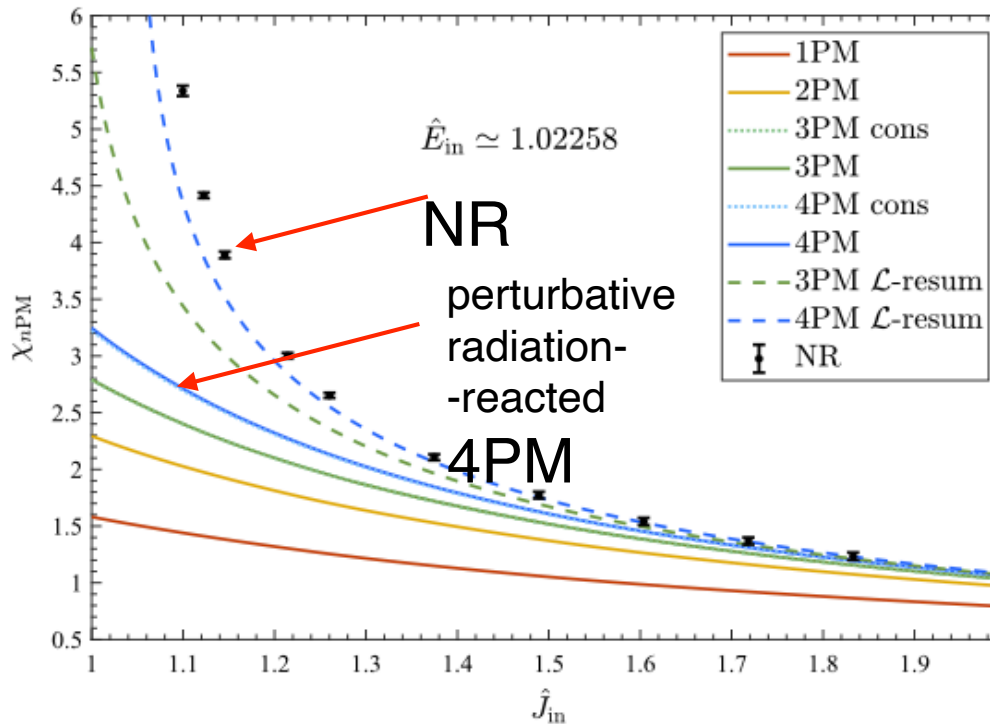
$$\chi_{n\text{PM}}(\gamma, j) \equiv \sum_{i=1}^n 2 \frac{\chi_i(\gamma)}{j^i} \quad \rightarrow \quad \mu^2 + g_{\text{eff}}^{\mu\nu} P_\mu P_\nu + Q(X^\mu, P_\mu) = 0,$$

$$p_{\bar{r}}^2 + \frac{j^2}{\bar{r}^2} = p_\infty^2 + w(\bar{r}, \gamma).$$

$$w(\bar{r}, \gamma) = \frac{w_1(\gamma)}{\bar{r}} + \frac{w_2(\gamma)}{\bar{r}^2} + \frac{w_3(\gamma)}{\bar{r}^3} + \frac{w_4(\gamma)}{\bar{r}^4} + O\left[\frac{1}{\bar{r}^5}\right]$$

Newtonianlike EOB radial potential

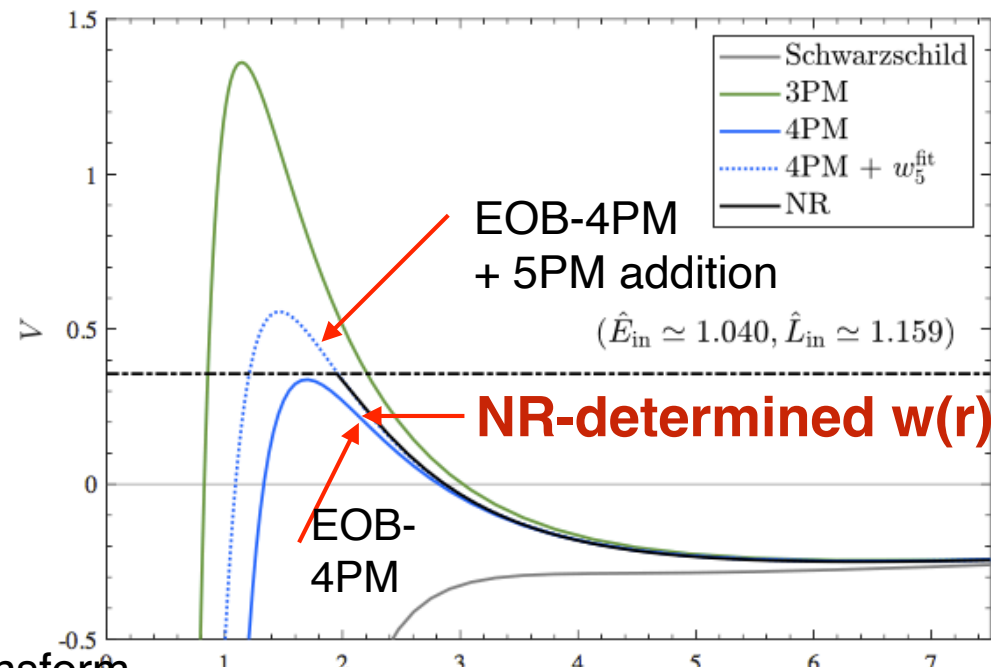
$$\chi_{n\text{PM}}^{w\text{eob}}(\gamma, j) \equiv 2j \int_0^{\bar{u}_{\text{max}}(\gamma, j)} \frac{d\bar{u}}{\sqrt{p_\infty^2 + w_{n\text{PM}}(\bar{u}, \gamma) - j^2 \bar{u}^2}} - \pi.$$



Strong-field scattering of two spinning black holes: Numerical Relativity versus post-Minkowskian gravity (Rettegno et al.'23)

Higher-energy non-spinning:

Comparison between the effective potential $V=L^2/r^2-w(r)$ extracted from NR simulations and its EOB-PM equivalent



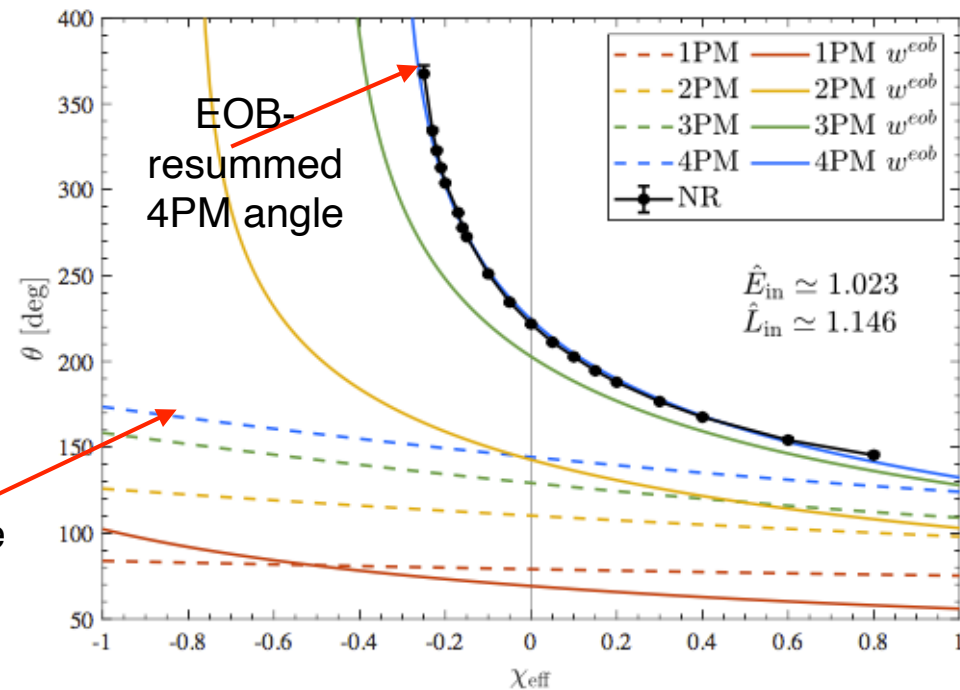
In the **spin-aligned case**, one can transform the PM-expanded scattering angle

$$\theta_{n\text{PM}}(\gamma, \ell, S_i) \equiv \sum_{k=1}^n 2 \frac{\theta_k(\gamma, \ell, S_i)}{\ell^k}$$

into an equivalent **spin-dependent EOB potential**

$$w_{n\text{PM}}(\bar{r}, \gamma, \ell, S_i) = w^{\text{orb}}(\bar{r}, \gamma) + \frac{\ell w_{n\text{PM}}^{\text{S}}(\bar{r}, \gamma)}{\bar{r}^2} + \frac{w_{n\text{PM}}^{\text{S}^2}(\bar{r}, \gamma)}{\bar{r}^2} + \frac{\ell w_{n\text{PM}}^{\text{S}^3}(\bar{r}, \gamma)}{\bar{r}^4} + \frac{w_{n\text{PM}}^{\text{S}^4}(\bar{r}, \gamma)}{\bar{r}^4}.$$

perturbative 4PM angle



PM waveform computation $W(k^\mu) = \epsilon^\mu \epsilon^\nu h_{\mu\nu}(\omega, \theta, \phi)$

$G^1=1$ PM (linearized, Einstein 1918) stationary $\propto \delta(\omega)$

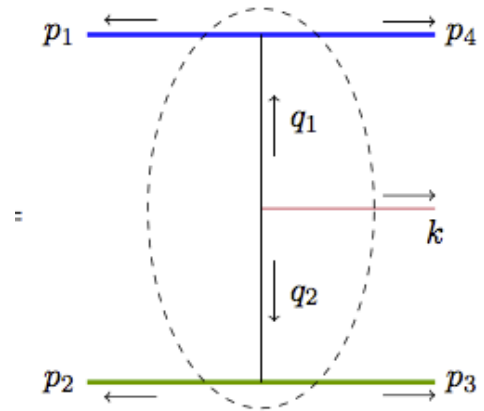
LO (tree level) waveform

$G^2=2$ PM: **classical time-domain $W(t, n)$** : Kovacs-Thorne 1977

quantum-based: yields $W(k, p_1, p_2, p_3, p_4) = W(k, p_1, p_2, q_1)$

Johansson-Ochirov'15, Goldberger-Ridgway'17 Luna-Nicholson-OConnell-White'18

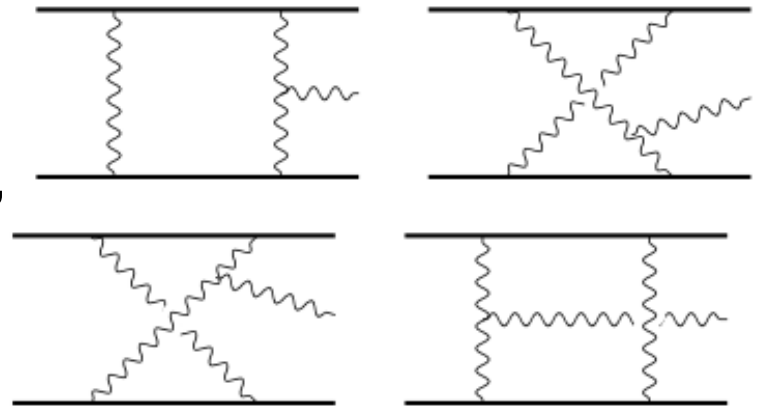
Mougiakakos-Riva-Vernizzi'21, Bautista-Siemonsen'22, De Angelis-Gonzo-Novichkov'23



NLO (one-loop) waveform

$G^3=3$ PM

Brandhuber+'23, Herderschee+'23, Georgoudis+'23,
Bohnenblust+'24



5-point HEFT one-loop amplitude

$\rightarrow O(G^3)$ waveform via KMOC

**5-point
amplitude:
2 $\rightarrow$ 3**

$$\mathcal{M}(\epsilon, k, p_1, p_2, q_1, q_2)$$

$$\equiv i \langle p_3 p_4 | \hat{a}(k) \mathbb{T} | p_1 p_2 \rangle + \langle p_3 p_4 | \mathbb{T}^\dagger \hat{a}(k) \mathbb{T} | p_1 p_2 \rangle$$

$$= i \langle p_3 p_4 k | \mathbb{T} | p_1 p_2 \rangle + \langle p_3 p_4 | \mathbb{T}^\dagger \hat{a}(k) \mathbb{T} | p_1 p_2 \rangle,$$

$$\mathcal{W}(\epsilon, k, p_1, p_2, b_1, b_2)$$

$$= \frac{2i}{\kappa} \int \mu(k, q_1, q_2) e^{-i(q_1 \cdot b_1 + q_2 \cdot b_2)} \mathcal{M}(\epsilon, k, p_1, p_2, q_1, q_2),$$

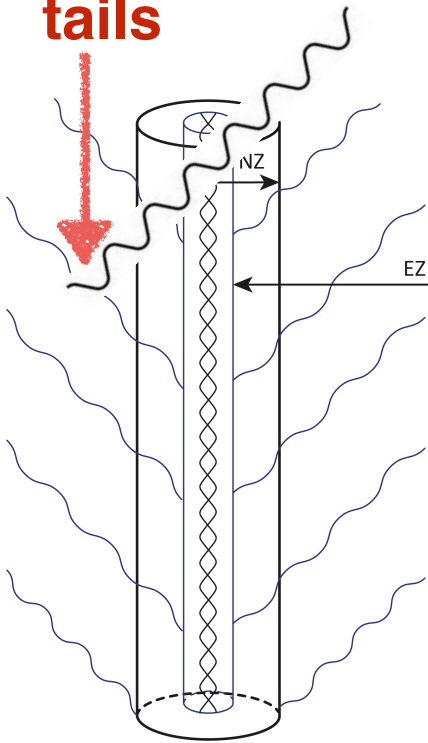
important « cut term » (Caron-Huot+'23)

Fourier transform over q_1, q_2

Comparing one-loop amplitude to MPM waveform

(Bini-TD-Geralico'23)

tails



algorithmic

STF tensors encoding multipole moments (related to the source moments I_L, J_L)

$$\begin{aligned}
 g &= \eta + Gh_1 + G^2h_2 + G^3h_3 + \dots, \\
 \square h_1 &= 0, \\
 \square h_2 &= \partial\partial h_1 h_1, \\
 \square h_3 &= \partial\partial h_1 h_1 h_1 + \partial\partial h_1 h_2, \\
 h_1 &= \sum_{\ell} \partial_{i_1 i_2 \dots i_{\ell}} \left(\frac{M_{i_1 i_2 \dots i_{\ell}}(t - r/c)}{r} \right) + \partial\partial \dots \partial \left(\frac{\epsilon_{j_1 j_2 k} S_{k j_3 \dots j_{\ell}}(t - r/c)}{r} \right), \\
 h_2 &= FP_B \square_{\text{ret}}^{-1} \left(\left(\frac{r}{r_0} \right)^B \partial\partial h_1 h_1 \right) + \dots, \\
 h_3 &= FP_B \square_{\text{ret}}^{-1} \dots
 \end{aligned}$$

radiative multipole moments (observable at infinity) U_L, V_L

$$rh_{ij}^{\text{TT}} = \frac{4G}{c^2} P(n)_{ijab} \sum_{l=2}^{\infty} \frac{1}{c^l} \frac{1}{l!} \left(U_{abL-2} n_{L-2} - \frac{2l}{c(l+1)} n_{cL-2} \epsilon_{cd(a} V_{b)dL-2} \right)$$

$$\mathcal{W}^{\text{MPM}}(k, b, u_1, u_2, m_1, m_2) = -i \frac{\kappa}{2} \epsilon^{\mu} \epsilon^{\nu} h_{\mu\nu}^{\text{MPM}}(\omega, \theta, \phi) = -i \frac{\kappa}{2} \int dt e^{i\omega t} \epsilon^{\mu} \epsilon^{\nu} h_{\mu\nu}^{\text{MPM}}(t, \theta, \phi)$$

$$\mathcal{W}^{\text{HEFT}}(k, b, u_1, u_2, m_1, m_2) =$$

1d Fourier transform of MPM

$$e^{i \frac{b_1 + b_2}{2} \cdot k} \int \frac{d^D q}{(2\pi)^{D-2}} \delta\left(2p_1 \cdot \left(q + \frac{k}{2}\right)\right) \delta\left(2p_2 \cdot \left(-q + \frac{k}{2}\right)\right) e^{iq \cdot (b_1 - b_2)} \mathcal{M}_{5, \text{HEFT}}^{(1)}\left(q + \frac{k}{2}, -q + \frac{k}{2}; h\right)$$

Comparison one-loop vs MPM waveform **at 2.5PN**

$$W(t, \theta, \phi) \sim \frac{1}{c^4} \left(\underbrace{G \text{ (stationary)}}_{\text{tree-level}} + \underbrace{G^2 \left(1 + \frac{1}{c^1} + \frac{1}{c^2} + \frac{1}{c^3} + \dots\right)}_{\text{one-loop}} + G^3 \left(1 + \frac{1}{c^1} + \frac{1}{c^2} + \frac{1}{c^3} + \dots\right) + O(G^4) \right)$$

Aim: accuracy up to radiation-reaction effects: $O(1/c^5)$ beyond LO quadrupole

$$U_{ij}(\omega) \sim \left(G \left(1 + \frac{1}{c^2} + \frac{1}{c^4}\right) + G^2 \left(1 + \frac{1}{c^2} + \frac{1}{c^3} + \frac{1}{c^4} + \frac{1}{c^5}\right) + O(G^4) \right) + O\left(\frac{1}{c^6}\right)$$

Newtonian G^2

LO tail

rad-reac plus
similar effects

$$U_{ij}^{\text{tail}}(t) = \frac{2G\mathcal{M}}{c^3} \int_0^\infty d\tau I_{ij}^{(4)}(t - \tau) \left(\ln\left(\frac{\tau}{2b_0}\right) + \frac{11}{12} \right)$$

Main results of the **initial EFT-MPM comparison** (Bini-TD-Geralico, 2023):

mismatch at the Newtonian level, except if one refers the one-loop amp. to classical averaged momenta, rather than incoming momenta; **then** the terms linked to time-even PN corrections to multipoles **agree** but there are **many mismatches at the G^2/c^5 level**

Updated comparisons (Georgoudis et al.'23,'24, Bini et al. '24) lead to **perfect agreement** after taking into account three subtle effects:

- (1) the bilinear-in-amplitude KMOC term generates the needed rotation
- (2) IR divergences generate an additional **$(D-4)/(D-4)$** contribution
- (3) **zero-frequency gravitons** contribute additional terms at $\hbar \sim G$ and $\hbar \sim G^3$
- (4) interesting links between zero-freq gravitons and BMS frame (Veneziano-Vilkovisky)

Quadrupolar bremsstrahlung waveform at the **3.5PN** accuracy

(Bini,TD, Geralico, arXiv:2604.21522 [gr-qc])

Extending the MPM accuracy to 3.5PN and to the 2-loop level:
i.e. $O(G^4)$ contributions to $h(\omega)$, or $O(G^3)$ contribs to $W(\omega)=h/(4 G)$

Main motivation: high-precision MPM bremsstrahlung waveform
at **2-loop** to provide future comparison

Surprises: added MPM complexity of reaching the 3.5PN level,
and an interesting finding about **BMS**!

$$\begin{aligned} h_c(t_r, \theta, \phi) &= \lim_{R \rightarrow \infty} (R(h_+ - i h_\times)) \\ &= \lim_{R \rightarrow \infty} \bar{m}^\mu \bar{m}^\nu R h_{\mu\nu} . \end{aligned}$$

$$W(t_r, \theta, \phi) \equiv \frac{c^4}{4G} h_c(t_r, \theta, \phi)$$

$$t_r = t - \frac{r}{c} - \frac{2G\mathcal{M}}{c^3} \ln \frac{r}{b_0} ,$$

$$\hat{W}(\omega, \theta, \phi) = \int_{-\infty}^{\infty} dt_r e^{i\omega t_r} W(t_r, \theta, \phi)$$

Focus on the quadrupolar waveform $l=2$ $\hat{U}_2(\omega, \theta, \phi) = \frac{1}{2!} \bar{m}^i \bar{m}^j \hat{U}_{ij}(\omega)$

radiative moment at infinity $\nearrow$

$$U_{ij} = M_{ij}^{(2)} + \eta^3 \overline{U_{ij}^{1.5\text{PN}}} + \eta^5 \overline{U_{ij}^{2.5\text{PN}}} + \eta^6 \overline{U_{ij}^{3\text{PN}}} + \eta^7 \overline{U_{ij}^{3.5\text{PN}}} + O(\eta^8)$$

source moment $\nwarrow$ extra contributions from gauge moments, e.g: $\swarrow$

$$M_{ij} = I_{ij} + 4G\eta^5 M_{ij}^{W_0 I_2} + 4G\eta^7 [M_{ij}^{W_2 I_2} + M_{ij}^{Y_2 I_2} + M_{ij}^{X_0 I_2} + M_{ij}^{W_1 I_3} + M_{ij}^{Y_1 I_3} + M_{ij}^{W_0 W_2} + M_{ij}^{W_1 W_1} + M_{ij}^{W_0 Y_2} + M_{ij}^{W_1 Y_1} + M_{ij}^{Z_1 I_2} + M_{ij}^{W_1 J_2} + M_{ij}^{Y_1 J_2}], \quad (3.5)$$

$$M_{ij}^{W_0 I_2} = W^{(2)} I_{ij} - W^{(1)} I_{ij}^{(1)},$$

$$M_{ij}^{W_2 I_2} = \frac{4}{7} W_{a\langle i}^{(1)} I_{j\rangle a}^{(3)} + \frac{6}{7} W_{a\langle i} I_{j\rangle a}^{(4)}$$

hereditary terms:
tails,
memory,
instantan.

$$\overline{U_{ij}^{1.5\text{PN}}}(t_r) = 2G\mathcal{M}\eta^3 \int_0^{+\infty} d\tau M_{ij}^{(4)}(t_r - \tau) \left[\ln\left(\frac{c\tau}{2b_0}\right) + \frac{11}{12} \right],$$

$$\overline{U_{ij}^{2.5\text{PN}}}(t_r) = G\eta^5 \left\{ \overline{U_{ij}^{2.5\text{PN}}(\text{mem})} + \overline{U_{ij}^{2.5\text{PN}} I_2 I_2} + \overline{U_{ij}^{2.5\text{PN}} I_2 J_1} \right\},$$

$$\overline{U_{ij}^{2.5\text{PN}}(\text{mem})}(t_r) = -\frac{2}{7} \int_0^{+\infty} d\tau [M_{a\langle i}^{(3)} M_{j\rangle a}^{(3)}](t_r - \tau),$$

$$\overline{U_{ij}^{2.5\text{PN}} I_2 I_2}(t_r) = \frac{1}{7} M_{a\langle i}^{(5)} M_{j\rangle a} - \frac{5}{7} M_{a\langle i}^{(4)} M_{j\rangle a}^{(1)} - \frac{2}{7} M_{a\langle i}^{(3)} M_{j\rangle a}^{(2)},$$

$$\overline{U_{ij}^{2.5\text{PN}} I_2 J_1}(t_r) = \frac{1}{3} \epsilon_{ab\langle i} M_{j\rangle a}^{(4)} S_b,$$

$$\overline{U_{ij}^{3\text{PN}}}(t_r) = 2G^2 \mathcal{M}^2 \eta^6 \int_0^{+\infty} d\tau M_{ij}^{(5)}(t_r - \tau) \left[\ln^2\left(\frac{c\tau}{2b_0}\right) + \frac{11}{6} \ln\left(\frac{c\tau}{2b_0}\right) - \frac{107}{105} \ln\left(\frac{c\tau}{2r_0}\right) + \frac{124627}{44100} \right]$$

tails of tails $\nwarrow$

$$U_{ij}^{\overline{3.5\text{PN}}} = G\eta^7 \left[U_{ij}^{\overline{3.5\text{PN}}(\text{mem})} + U_{ij}^{\overline{3.5\text{PN}}I_2I_4} + U_{ij}^{\overline{3.5\text{PN}}I_3I_3} + U_{ij}^{\overline{3.5\text{PN}}J_1J_3} + U_{ij}^{\overline{3.5\text{PN}}J_2J_2} + U_{ij}^{\overline{3.5\text{PN}}J_3I_2} + U_{ij}^{\overline{3.5\text{P}}} \right]$$

$$U_{ij}^{\overline{3.5\text{PN}}(\text{mem})} = \int_0^{+\infty} d\tau \left[-\frac{5}{756} M_{ab}^{(4)} M_{ijab}^{(4)} - \frac{32}{63} S_{a\langle i}^{(3)} S_{j\rangle a}^{(3)} \right] (t_r - \tau) \\ + \epsilon_{ac\langle i} \int_0^{+\infty} d\tau \left[\frac{5}{42} S_{j\rangle cb}^{(4)} M_{ab}^{(3)} - \frac{20}{189} M_{j\rangle cb}^{(4)} S_{ab}^{(3)} \right] (t_r - \tau),$$

$$U_{ij}^{\overline{3.5\text{PN}}I_2I_4} = -\frac{1}{432} M_{ab} M_{ijab}^{(7)} + \frac{1}{432} M_{ab}^{(1)} M_{ijab}^{(6)} - \frac{5}{756} M_{ab}^{(2)} M_{ijab}^{(5)} + \frac{19}{648} M_{ab}^{(3)} M_{ijab}^{(4)} + \frac{1957}{3024} M_{ab}^{(4)} M_{ijab}^{(3)} \\ + \frac{1685}{1008} M_{ab}^{(5)} M_{ijab}^{(2)} + \frac{41}{28} M_{ab}^{(6)} M_{ijab}^{(1)} + \frac{91}{216} M_{ab}^{(7)} M_{ijab},$$

$$U_{ij}^{\overline{3.5\text{PN}}I_3I_3} = -\frac{5}{252} M_{ab\langle i} M_{j\rangle ab}^{(7)} + \frac{5}{189} M_{ab\langle i}^{(1)} M_{j\rangle ab}^{(6)} + \frac{5}{126} M_{ab\langle i}^{(2)} M_{j\rangle ab}^{(5)} + \frac{5}{2268} M_{ab\langle i}^{(3)} M_{j\rangle ab}^{(4)},$$

$$U_{ij}^{\overline{3.5\text{PN}}J_1J_3} = \frac{5}{42} S_a S_{ija}^{(5)},$$

$$U_{ij}^{\overline{3.5\text{PN}}J_2J_2} = \frac{80}{63} S_{a\langle i} S_{j\rangle a}^{(5)} + \frac{16}{63} S_{a\langle i}^{(1)} S_{j\rangle a}^{(4)} - \frac{64}{63} S_{a\langle i}^{(2)} S_{j\rangle a}^{(3)},$$

$$U_{ij}^{\overline{3.5\text{PN}}J_3I_2} = \epsilon_{ac\langle i} \left(\frac{1}{168} S_{j\rangle bc}^{(6)} M_{ab} + \frac{1}{24} S_{j\rangle bc}^{(5)} M_{ab}^{(1)} + \frac{1}{28} S_{j\rangle bc}^{(4)} M_{ab}^{(2)} - \frac{1}{6} S_{j\rangle bc}^{(3)} M_{ab}^{(3)} + \frac{3}{56} S_{j\rangle bc}^{(2)} M_{ab}^{(4)} \right. \\ \left. + \frac{187}{168} S_{j\rangle bc}^{(1)} M_{ab}^{(5)} + \frac{65}{84} S_{j\rangle bc} M_{ab}^{(6)} \right),$$

$$U_{ij}^{\overline{3.5\text{PN}}I_3J_2} = \epsilon_{ac\langle i} \left(\frac{1}{189} M_{j\rangle bc}^{(6)} S_{ab} - \frac{1}{189} M_{j\rangle bc}^{(5)} S_{ab}^{(1)} + \frac{10}{189} M_{j\rangle bc}^{(4)} S_{ab}^{(2)} + \frac{32}{189} M_{j\rangle bc}^{(3)} S_{ab}^{(3)} + \frac{65}{189} M_{j\rangle bc}^{(2)} S_{ab}^{(4)} \right. \\ \left. - \frac{5}{189} M_{j\rangle bc}^{(1)} S_{ab}^{(5)} - \frac{10}{63} M_{j\rangle bc} S_{ab}^{(6)} \right).$$

in addition to the MPM nonlinear contributions one must explicitly solve the
 3.5PN-accurate hyperbolic motion, including rad-reac effects,
 and finally compute the Fourier transform from $t_r \rightarrow \omega$

e.g.

$$\begin{aligned} \delta^{\text{rr}, G^3} x = & \left(-\frac{8}{5} \frac{1}{(1+T^2)^{3/2}} \text{arcsinh}(T) - \frac{47}{15} \arctan(T) - \frac{47}{30} \pi \right. \\ & \left. - \frac{8}{5} (1+T^2)^{1/2} + \frac{8}{5(1+T^2)^{1/2}} - \frac{8}{5} T - \frac{59}{15(1+T^2)} T \right) \frac{G^3 M^3 \nu}{p_\infty b^2} \eta^5 \\ & + \left[\left(\left(\frac{464}{105} + \frac{4}{5} \nu \right) \frac{1}{(1+T^2)^{3/2}} + \left(-\frac{12}{5} \nu - \frac{98}{5} \right) \frac{1}{(1+T^2)^{5/2}} + \frac{16}{(1+T^2)^{7/2}} \right) \text{arcsinh}(T) \right. \\ & + \left(-\frac{172}{35} - \frac{12}{5} \nu \right) (1+T^2)^{1/2} - \frac{172}{35} T - \frac{4997}{1680} \pi - \frac{4997}{840} \arctan(T) \\ & - \frac{12}{5} \nu T + \frac{40}{7(1+T^2)^{1/2}} + \left(-\frac{2357}{840} - 2\nu \right) \frac{T}{(1+T^2)} + \left(\frac{12}{5} \nu - \frac{4}{5} \right) \frac{1}{(1+T^2)^{3/2}} \\ & \left. + \left(-\frac{47}{5} \nu - \frac{97}{12} \right) \frac{T}{(1+T^2)^2} + \frac{16T}{(1+T^2)^3} \right] \frac{\nu G^3 M^3 p_\infty}{b^2} \eta^7, \end{aligned}$$

« iterated Bessel functions »
 satisfying
 $\text{BesselOperator}[Q] = u^a \text{BesselK}$

at 2-loop

$$\begin{aligned} Q_1^{\text{as}}(u) &\equiv \int_{-\infty}^{+\infty} dT e^{iuT} \frac{\text{arcsinh}(T)}{(1+T^2)}, \\ Q_{1/2}^{\text{as2}}(u) &\equiv \int_{-\infty}^{+\infty} dT e^{iuT} \frac{\text{arcsinh}^2(T)}{(1+T^2)^{1/2}}, \\ Q_{1/2}^{\text{at}}(u) &\equiv \int_{-\infty}^{+\infty} dT e^{iuT} \frac{\arctan(T)}{(1+T^2)^{1/2}}. \end{aligned}$$

sample **two-loop** results, involving:

$$K_0(u), K_1(u), K_{1/2}(u), Q_1^{\text{as}}(u), Q_{1/2}^{\text{as}2}(u), Q_{1/2}^{\text{at}}(u)$$

G^3 eta^0

$$\begin{array}{ll} \overline{U_2^{\bar{m}_1^2 G^3 \eta^0}} & \frac{\nu}{b^2 p_\infty^5} \left[(2 + u^2) K_0(u) + 2u K_1(u) - \frac{1}{2} u^3 Q_{1/2}^{\text{as}2'}(u) \right] \\ U_2^{\bar{m}_1 \bar{m}_2 G^3 \eta^0} & \frac{\nu}{b^2 p_\infty^5} \left[6iu K_0(u) + (4i + 2iu^2) K_1(u) + iu^3 Q_{1/2}^{\text{as}2}(u) - iu^2 Q_{1/2}^{\text{as}2'}(u) \right] \\ U_2^{\bar{m}_2^2 G^3 \eta^0} & \frac{\nu}{b^2 p_\infty^5} \left[(-2 - u^2) K_0(u) - 3u K_1(u) - \frac{1}{2} u^2 Q_{1/2}^{\text{as}2}(u) + \frac{1}{2} u^3 Q_{1/2}^{\text{as}2'}(u) \right] \end{array}$$

G^3 eta^7

$$\begin{array}{ll} \overline{U_2^{\bar{m}_1^2 G^3 \eta^7}} & \frac{\nu p_\infty^2}{b^2} [A_{K_0}^{\bar{m}_1^2} K_0(u) + A_{K_1}^{\bar{m}_1^2} K_1(u) + A_{e^{-u}}^{\bar{m}_1^2} e^{-u}] \\ A_{K_0}^{\bar{m}_1^2} & \left(\frac{32\nu^2}{7} - \frac{1472\nu}{105} \right) u^2 + \left(\left(-\frac{\pi^2}{7} - \frac{3853i\pi}{630} \right) \nu^2 + \left(-\frac{59\pi^2}{42} + \frac{1117i\pi}{180} \right) \nu + \frac{47\pi^2}{84} - \frac{517i\pi}{504} \right) u^4 + \mathcal{L} \left(-\frac{2}{7} i\pi \nu^2 - \frac{59i\pi\nu}{21} + \frac{47i\pi}{42} \right) u^4 \\ A_{K_1}^{\bar{m}_1^2} & \left(\left(-\frac{41\pi^2}{189} + \frac{1592i\pi}{567} \right) \nu^2 + \left(\frac{247\pi^2}{378} - \frac{16523i\pi}{5670} \right) \nu - \frac{71\pi^2}{378} + \frac{781i\pi}{2268} \right) u^5 \\ & \left(\left(-\frac{96}{35} + \frac{386i\pi}{945} - \frac{8\pi^2}{63} \right) \nu^2 + \left(\frac{152}{105} - \frac{541i\pi}{2520} - \frac{31\pi^2}{252} \right) \nu + \frac{2071\pi^2}{504} - \frac{22781i\pi}{3024} \right) u^3 \\ & + \mathcal{L} \left(\left(-\frac{82}{189} i\pi \nu^2 + \frac{247i\pi\nu}{189} - \frac{71i\pi}{189} \right) u^5 + \left(-\frac{16}{63} i\pi \nu^2 - \frac{31i\pi\nu}{126} + \frac{2071i\pi}{252} \right) u^3 \right) \\ A_{e^{-u}}^{\bar{m}_1^2} & -\frac{409i\pi\nu^2}{1344} - \frac{3397i\pi\nu}{2688} + \left(-\frac{16073i\pi\nu^2}{12096} + \left(-\frac{39\pi^2}{14} + \frac{114349i\pi}{17280} \right) \nu + \frac{61\pi^2}{28} - \frac{671i\pi}{168} \right) u^3 \\ & + \left(\frac{14221i\pi\nu^2}{10080} + \left(-\frac{33\pi^2}{14} - \frac{84703i\pi}{60480} \right) \nu - \frac{97\pi^2}{28} + \frac{1067i\pi}{168} \right) u^2 + \left(-\frac{477}{320} i\pi \nu^2 + \left(-\frac{33\pi^2}{14} + \frac{13939i\pi}{2688} \right) \nu - \frac{97\pi^2}{28} + \frac{1067i\pi}{168} \right) u \\ & + \frac{-\frac{409i\pi\nu^2}{1344} - \frac{3397i\pi\nu}{2688}}{u} + \mathcal{L} \left(\left(\frac{61i\pi}{14} - \frac{39i\pi\nu}{7} \right) u^3 + \left(-\frac{33}{7} i\pi\nu - \frac{97i\pi}{14} \right) u^2 + \left(-\frac{33}{7} i\pi\nu - \frac{97i\pi}{14} \right) u \right) \end{array}$$

Comparison with 3.5PN-accurate one-loop waveform

(Heissenberg-Russo 25, De Angelis, 25)

Besides checks from log-dependence, SF (against Geralico'26), and soft limit-memory term (including nonlinear memory)

$$\delta U_2 = U_2^{\text{EFT}} - U_2^{\text{MPM}}$$

using
 $\ln(2b_0\mu_{\text{IR}}) + \gamma_E = 0$

we found the following **mismatches**

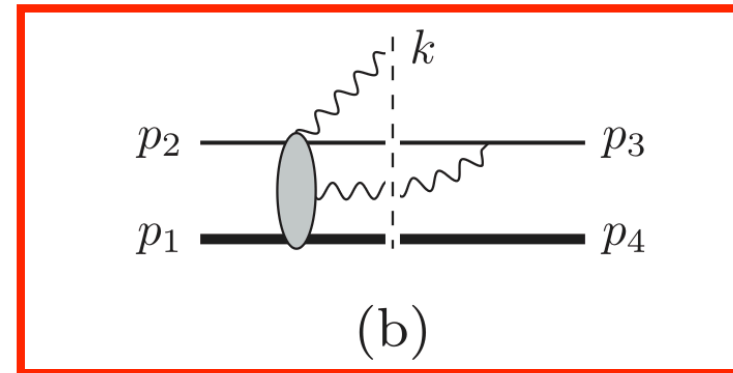
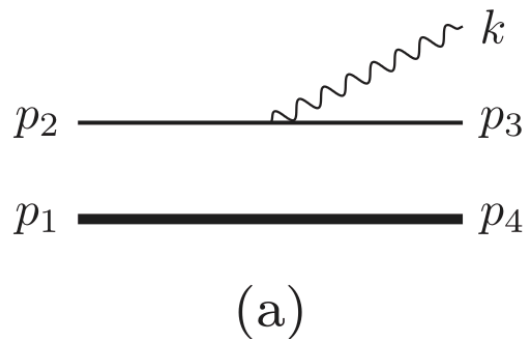
$$\begin{aligned}\delta U_{22} &= i\nu^2 G^2 \sqrt{\frac{\pi}{5}} p_\infty^4 \frac{4}{105b} (1 - 4\nu) u [2(3 + 2u + 5u^2) K_0(u) \\ &\quad + (16 + 9u + 10u^2) K_1(u)], \\ \delta U_{20} &= i\nu^2 G^2 \sqrt{\frac{2\pi}{15}} p_\infty^4 \frac{4}{35b} (1 - 4\nu) u [2K_0(u) \\ &\quad + 11u K_1(u)] \\ \delta U_{2\bar{2}} &= i\nu^2 G^2 \sqrt{\frac{\pi}{5}} p_\infty^4 \frac{4}{105b} (1 - 4\nu) u [2(3 - 2u + 5u^2) K_0(u) \\ &\quad + (-16 + 9u - 10u^2) K_1(u)]\end{aligned}\tag{9.10}$$

Structure of the one-loop waveform (from Tale of Two Formalism paper)

(D-4)/(D-4) term !

$$W^{\text{EFT}}(\omega, \mathbf{n}) = W^{G^1} + W^{G^2(0)} + W^{G^2\epsilon/\epsilon} \\ + W^{G^2\text{cut}} + W^{G^2\text{disc,reg}'} + \delta W^{G^2\text{rot}}$$

delicate term linked to **disconnected diagrams involving zero-energy gravitons**



$$\mathcal{M}_{\text{disc}} = -i\kappa^2 \mathcal{M}^{\text{tree}} [m_2^2 w_2^2 I(u_3, k) + m_1^2 w_1^2 I(u_4, k)],$$

$$I(u_3, k) = \frac{1}{m_2} \int \frac{d^d \ell}{(2\pi)^d} \frac{\hat{\delta}(2u_3 \cdot \ell) \hat{\delta}(\ell^2) \Theta(\ell^0)}{2\ell \cdot k} \\ = \frac{1}{m_2 \omega} \int \frac{d^d \ell}{(2\pi)^d} \frac{\hat{\delta}(2u_3 \cdot \ell) \hat{\delta}(\ell^2) \Theta(\ell^0)}{2\ell \cdot \hat{k}}$$

logarithmically divergent
integral which needed
an additional regularization
besides dim-reg

$$\mathcal{M}_{\text{disc}} = \mathcal{S}^{\text{1 loop disc}} - i\omega GE \left[\frac{1}{\epsilon} - \log \frac{\beta^2}{\pi} \right] \mathcal{M}^{\text{tree}},$$

$$\mathcal{S}^{\text{1 loop disc}} = iG \left[m_1 w_1 \log \frac{w_1^2}{\omega^2} + m_2 w_2 \log \frac{w_2^2}{\omega^2} \right] \mathcal{M}^{\text{tree}}$$

$$= +i\omega \beta^{\text{VV}}(\theta, \phi) \mathcal{M}^{\text{tree}}$$

yields the **Veneziano-Vilkovisky supertranslation** aligning the EFT BMS frame with the MPM BMS frame

However, we found that

$$\delta U_2 = +i\omega \beta_{l=1}^{\text{VV}}(\theta, \phi) W_{\text{tree}}$$

ie one needs (at 3.5PN) to subtract the dipolar part of β^{VV} from the disconnected cut contribution (or add β_1^{VV} to W^{MPM})

$$W^{G^2 \text{disc, new}} = i\omega (\beta^{\text{VV}}(\theta, \phi) - \beta_1^{\text{VV}}(\theta, \phi) n^Y) W^{\text{tree}}$$

This is fully compatible with the fact that β^{VV} is, actually, only defined modulo $l=0$ and $l=1$ homogeneous contributions

Conclusions

There exist useful **synergies** between various approaches to the 2-body pb.

Scattering approaches have made impressive headway in few years, though they seem close to reaching their practical limits.

Comparisons between various approaches to the 2-body pb are important and allow one to understand new, subtle aspects of gravitational interactions

Though Numerical Relativity is and will remain very important and useful, analytical approaches will continue to play an important role.

For **GW generation** the « traditional » MPM+PN approach is the only one which has reached a (LIGO-relevant) high precision: 4PN for quasi-circular motions (4-loop) (Blanchet et al) and 3.5PN+ 2-loop for scattering motions

The many subtle effects linked to **radiation-reaction** and the decomposition « conservative + rad reac » remain to be fully clarified

The flexible analytical nature of the EOB formalism makes it useful for incorporating new information in LIGO-useful form.

